

SS 701:2023
(ICS 97.220.01, 97.220.40)

SINGAPORE STANDARD

Code of practice for inland and open water sporting activities



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Foreword

This Singapore Standard was prepared by the Working Group on Open Water Sport Safety set up by the Technical Committee on Workplace Safety and Health under the purview of the Safety and Quality Standards Committee.

The standard emphasises personal responsibility while ensuring that it also covers the responsibilities of stakeholders, the safety management system, the readiness of locations/facilities, and the adequacies of equipment supplied to the participant(s). Additionally, the standard highlights the open water safety requirements of the common inland and open water sporting activities (as shown in the Annexes) in Singapore.

In preparing this standard, reference was made to the following Singapore Standards:

- SS 681:2022 Code of practice for sport safety
- SS 700:2023 Code of practice for water safety – Aquatic facilities.

In addition, the Working Group engaged key stakeholders of open water sports to stipulate minimum requirements for the prevention of injuries and fatalities from open water activities (e.g. recreational, competitive, and educational) that are planned by the public and organised groups.

It is presupposed that in the course of their work, stakeholders will comply with all relevant regulatory and statutory requirements. Some examples of relevant regulations and acts are listed in the Bibliography. The Singapore Standards Council and Enterprise Singapore shall not be responsible for identifying all of such legal obligations.

Permission has also been sought from the following organisations for the reproduction of materials from their publications into this standard:

1. International Life Saving Federation (Medical Committee) Figure 1 from the Medical Position Statement – MPS 19.
2. International Organization for Standardization (ISO)
 - Definitions of Drowning and Competent person from 3.2 and 3.6 of ISO 20380:2017, “Public swimming pools – Computer vision systems for the detection of drowning accidents in swimming pools – Safety requirements and test methods”
 - Definition of Personal flotation device (PFD) from 3.1 of ISO 12402-2:2020, “Personal flotation devices – Part 2: Lifejackets, performance level 275 – Safety requirements”.
3. Singapore Dragon Boat Association, Figures H.1, H.2 and H.3.
4. Singapore Power Boat Association, Figure 2.
5. Sport Singapore, Figures R.1 and R.2 from the Safe waterskiing and wakeboarding guide

Acknowledgement is made for the use of information from the above publications.

Attention is drawn to the possibility that some of the elements of this Singapore Standard may be the subject of patent rights. Enterprise Singapore shall not be held responsible for identifying any or all such patent rights.

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NOTE

1. *Singapore Standards (SSs) and Technical References (TRs) are reviewed periodically to keep abreast of technical changes, technological developments and industry practices. The changes are documented through the issue of either amendments or revisions. Where SSs are deemed to be stable, i.e. no foreseeable changes in them, they will be classified as “mature standards”. Mature standards will not be subject to further review unless there are requests to review such standards.*
2. *An SS or TR is voluntary in nature except when it is made mandatory by a regulatory authority. It can also be cited in contracts making its application a business necessity. Users are advised to assess and determine whether the SS or TR is suitable for their intended use or purpose. If required, they should refer to the relevant professionals or experts for advice on the use of the document. Enterprise Singapore and the Singapore Standards Council shall not be liable for any damages whether directly or indirectly suffered by anyone or any organisation as a result of the use of any SS or TR. Although care has been taken to draft this standard, users are also advised to ensure that they apply the information after due diligence.*
3. *Compliance with a SS or TR does not exempt users from any legal obligations.*

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Code of practice for inland and open water sporting activities

1 Scope

1.1 This standard specifies requirements that are aimed at preventing injuries and fatalities from recreational, competitive, and educational activities planned by the public and organised groups in inland and open waters.

1.2 The objective of this standard is to provide requirements and recommendations for stakeholders to address any health and safety concerns during the use of open water facilities/activity locations and during organised activities/events in an open water environment.

1.3 For the purpose of this standard,

a) open water bodies cover:

- beach fronts,
- coastal areas,
- deep sea,
- floodplains,
- lakes,
- oceans,
- rivers,
- sea waters,
- lagoons, and
- wetlands.

b) inland water bodies cover:

- creeks,
- ditches,
- flood ways,
- monsoon water drains,
- ponds,
- quarries,
- reservoirs,
- streams, and
- waterways.

NOTE – In Singapore, access to quarries by the public is not allowed.

1.4 This standard does not cover swimming pools. For provisions for swimming pools, refer to SS 700:2023.

2 Normative references

The following referenced documents are indispensable for the application of this standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

SS 681:2022 Code of practice for sport safety

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3 Terms and definitions

For the purpose of this Singapore Standard, the following terms and definitions apply.

3.1 Basic lifesaving skills

The ability to save lives through the following:

- a) Prevention of accidents by having sufficient knowledge on water safety and taking responsibility for one's own safety as well as the safety of those under supervision.
- b) Personal survival (self-rescue) by achieving the relevant certification.
- c) Rescue of others (land rescue to water rescue in respect of difficulty level) by achieving the relevant lifesaving certification and First Aid with cardiopulmonary resuscitation (CPR)/automated external defibrillator (AED) certification.

Note 1 to entry: Longer distances and lengths of time are necessary for competency in different water environments.

Note 2 to entry: For an example of certification in b), refer to SwimSafer (Stage 1: 10 m, Stage 2: 25 m, Stage 3: 25 m, Bronze: 100 m, Silver: 50 m with timing, Gold: 100 m with timing).

Note 3 to entry: Examples of lifesaving certification in c) are Lifesaving 1: 100 m, Lifesaving 2: 150 m, Lifesaving 3: 200 m.

3.2 Basic swimming skills

The capacity to swim briefly and cover a short distance in the deep end of a swimming pool. In open waters, these skills necessitate a higher level of proficiency and comfort, particularly in challenging conditions such as rough waters, sea waves and water currents. Basic swimming skills are defined as the ability to do the following:

- a) Step or jump into water with complete submersion and return to the surface;
- b) Turn around/turn over and orientate to safety;
- c) Float or tread water for at least 50 s;
- d) Utilise breathing techniques while performing swim skills, including forward movement in the water;
- e) Swim to safety for at least a distance of 25 m (basic skill) or swim at least 100 m using relaxed, restful strokes (advanced skill);
- f) Exit the water; and
- g) Perform all the above skills while fully clothed.

Note 1 to entry: Longer distances and lengths of time are necessary for competency in different water environments.

3.3 Buddy system

A safe practice in which participants are paired or teamed up to ensure each other's safety.

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3.4 Buoy

Floating markers or objects that are positioned in the water to indicate the presence of hazards and guide participants along safe channels, areas or programme routes.

3.5 Coach/trainer

Person or entity in charge of training or instructing participants and imparting specific skills relating to open water sports or activities.

3.6 Competent person

Designated individual or entity who has acquired through training, qualifications or experience, or a combination of these, the knowledge and skills enabling that person to perform a specified task.

[SOURCE: ISO 20380:2017 Public swimming pools – Computer vision systems for the detection of drowning accidents in swimming pools – Safety requirements and test methods]

3.7 Drowning

Process of experiencing respiratory impairment from submersion/immersion in liquid.

Note 1 to entry: Outcomes are classified as death, morbidity and no morbidity.

[SOURCE: World Health Organization]

3.8 Lifeguard

Person employed by the facility owner, operator, manager, or event organiser to oversee beach and/or water participants, ensure their safety, and respond to emergencies. This person holds a current lifesaving certification and a Standard First Aid certification, including CPR and AED. Additional certifications related to open water rescue and lifesaving are encouraged.

Note 1 to entry: In Singapore, the lifesaving certification is the Bronze Cross Award issued by the Singapore Life Saving Society.

3.9 Non-swimmer

Person who has not undergone any formal swimming competency test. This person also lacks the ability of skills to swim (as specified in 3.2) and requires direct assistance from a competent swimmer (within arm's reach) when engaging in open water activities.

Note 1 to entry: A non-swimmer typically panics when experiencing difficulty in water, and tends to remain vertical in water and appears to be climbing a vertical ladder, referred to as "climbing the ladder", with minimal or non-supportive leg action. Additionally, the non-swimmer is likely to be not facing the shore and likely to submerge, and typically attempts to grab any object, including the rescuer.

3.10 No-swim zone

Any area that has been identified and cordoned off as hazardous and unsuitable for swimming.

3.11 Organiser

Person or entity in charge of planning and coordinating open water activities or events.

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3.12 Owner/occupier

Person or entity residing in or using a property as its owners or tenants to manage facilities and/or activities.

3.13 Parent/adult guardian/caregiver

Person in charge of providing constant and close supervision to participants in open water activities or events.

3.14 Participant

Individual involved in open water activities, which can include the use of watercraft (e.g. kayak, paddle board, dragon boat, etc.).

Note 1 to entry: The term “participant” can be used interchangeably with “trainee”, “swimmer”, “diver”, “athlete”, “rower”, “kayaker”, “sailor”, “boater”, etc.

3.15 Personal flotation device (PFD)

A garment or device which, when correctly worn and used in water, provides the participant with a specific amount of buoyancy which increases the likelihood of survival.

[SOURCE: ISO 12402-2:2020 Personal flotation devices – Part 2: Lifejackets, performance level 275 – Safety requirements]

3.16 Powered pleasure craft

A pleasure craft propelled by machinery that is intended for use (whether such use is for private use or commercial use) within the port exclusively for sport or pleasure purposes but does not include any craft which is used to carry passengers on sightseeing tours within the port for which each such passenger is charged a separate and distinct fare.

Note 1 to entry: “Private use”, in relation to a pleasure craft, means the use of the pleasure craft within the port by its owner or his guests for sport or pleasure purposes, without any such guest having to pay any fee or any other form of consideration.

[SOURCE: MPA Pleasure Craft Regulations, Part 1, Reg 2(1) and Part IV, Reg 27]

3.17 Pre-participation questionnaire

A self-administered assessment completed by participants to evaluate their physical well-being before engaging in physical activities.

Note 1 to entry: Refer to the Sport Singapore website for an example of a pre-participation questionnaire.

3.18 Risk management

The continuous process of assessing risks and applying control measures systematically to minimise risk using practical and cost-effective methods.

3.19 Spotter

Person stationed on the shore or in the open water to watch for weaker swimmers, swimmers in distress, and participants who have fallen overboard.

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3.20 Stakeholder

Individual involved in or having an interest in open water activities such as an owner, occupier, organiser, vendor, lifeguard, coach, trainer, parent, adult guardian, caregiver, or participant.

3.21 Surface marker buoy (SMB)

A device that floats on the surface for the duration of the dive and is connected by a nylon line to a reel that is attached to the diver. Also known as a dive float or blob.

3.22 Weak swimmer

Person who has received limited swimming instruction, lacks swim proficiency and typically requires assistance from a competent swimmer (within arm's reach) during open water activities.

Note 1 to entry: A weak swimmer can use arms and legs for support. The weak swimmer is typically angled in the water (e.g. at approximately 45 °) and likely to grab the rescuer or a floatation aid. The head position can be tilted up and back, with the head turned towards safety or for help.

3.23 Use of “shall”, “should”, “may”, and “can”

In this standard, the following verbal forms are used:

- a) “shall” indicates that the requirement is strictly to be followed in order to conform to the standard and from which no deviation is permitted.
- b) “should” indicates a recommendation;
- c) “may” indicates a permission;
- d) “can” indicates a possibility or a capacity.

4 Stakeholders' responsibilities

4.1 Owner/occupier

4.1.1 Owners/occupiers of facilities that are used for their operations shall ensure the following:

- a) Plan and specify permissible activities, including control and/or limit access;
- b) Perform risk assessments and implement a risk management programme;
- c) Manage and monitor health and safety hazards at the facilities and during activities;
- d) Establish emergency action plans, and provide working emergency equipment and trained personnel;
- e) Disseminate essential information to all individuals, including permissible activities, demarcated areas, appropriate attire to be worn, health hazards, emergency equipment available, and safe and sheltered areas;
- f) Report and document major incidents, including drownings, to management and relevant government authorities for analysis and improvement actions; and
- g) Conduct routine equipment checks and maintenance in accordance with their specific use.

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4.1.2 Owners/occupiers of facilities should use the venue and equipment checklist in Annex A when conducting routine checks on the venue and equipment.

4.2 Organiser

4.2.1 Organisers of activities shall ensure the following:

- a) Plan and schedule water activities to prevent incompatible activities, for example by cordoning off and designating separate areas for swimming from areas for motorised boats;
- b) Segment participants by proficiency using methods like colour coding and stagger start times to prevent overcrowding;
- c) Plan for and provide emergency medical response in case of incidents;
- d) Conduct risk assessments, implement a risk management programme and adhere to location/facility rules and regulations;
- e) Disseminate safety information to all participants through various communication channels like leaflets, webpage, on-site briefings, orientation courses, etc.; and
- f) Provide necessary supervision, including safety boats, spotters, lifeguards, safety/emergency equipment, and trained personnel. They shall maintain line of sight with participants during the open water activities/events.

4.2.2 Organisers of open water sport activities or those bringing groups overseas for open water sports shall conduct a risk assessment prior to the trip. If possible, organisers should also carry out a site reconnaissance at the activity location. They shall also have in place an emergency response plan, including incident management and communication plans.

4.2.3 Organisers of activities should use the checklist for safe water activities in Annex B before the commencement of the activities.

4.3 Lifeguard

4.3.1 Lifeguards play an important role in observing and supervising participants' safety and carrying out emergency operations when required. They shall ensure the following:

- a) Maintain critical operations information, including activity schedules, layout plans, water conditions and evacuation routes;
- b) Implement shift rotations if supervision is for long durations;
- c) Clearly demarcate non-swimming areas, and display warnings on signboards, flags, buoys and other forms of static display;
- d) Establish prominent access control (ingress and egress) to the open water areas; and
- e) Monitor and guide participants on the safe use of the facilities, warn participants of hazards and dangerous situations and provide emergency services where necessary, reporting all incidents.

4.3.2 Lifeguards should wear easily identifiable uniforms, wide-brimmed hats, sunglasses, and use sunblock to protect themselves during prolonged outdoor exposure.

4.3.3 Lifeguards shall be deployed, and where practicable, be equipped with the following equipment:

- a) Communication – air horns, flare guns, mobile phones, walkie talkies, and whistles;

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- b) Medical – first aid kits, AEDs, oxygen devices, stretchers, cover litters, blankets, spinal boards, and head and neck immobilisation devices;
- c) Rescue – Rescue floatation devices (RFD) such as rescue tubes, binoculars, masks, snorkels, swim fins, rescue boats, jet skis and safety crafts (boats or kayaks); and
- d) Surveillance – Binoculars.

4.4 Participant

Participants shall ensure the following:

- a) Be aware of their own health status and/or undergo a pre-participation screening if they have not regularly participated in any sporting activities or have not undergone any health assessment in the last 2 years;
- b) Familiarise themselves with vital information provided by owners and operators regarding open water activities, events, locations, and facilities such as permissible activities, appropriate attire, possible health hazards, use of emergency equipment, etc.;
- c) Abide by the rules and regulations of the activity, event, location, or facility;
- d) Minimally know how to swim for purpose of self-preservation;
- e) Perform activities according to one's capabilities and intensity levels, progressing gradually;
- f) Wear necessary personal protective equipment such as a life vests (inflatable or non-inflatable), helmet and appropriate footwear;
- g) Avoid consumption of alcohol and medication with adverse effects (e.g. drowsiness) before any open water activity;
- h) Avoid touching or handling marine flora or fauna, unless trained to do so;
- i) Help look out for other participants; and
- j) Report any major or drowning incidents to the event organiser, facility owner, or occupier, and the relevant local authority.

4.5 Coach/trainer

4.5.1 Coaches and trainers shall possess current and valid certifications of the relevant open water activities (e.g. swimming, kayaking, rowing, sailing, boating, etc.). These include the following:

- a) Valid technical skill certifications recognised by the local relevant authority to conduct the type and level of activity provided for participants;
- b) Valid Standard First Aid (SFA) Certificate; and
- c) CPR + AED Certificate.

4.5.2 Coaches and trainers should hold relevant additional certifications for open water activities. When such certifications (e.g. open water lifesaving certifications such as Bronze Cross or higher), are lacking, additional risk mitigation measures should be implemented (e.g. engaging a lifeguard, safety boat or ambulance).

NOTE 1 – Refer to the Lifeguard Singapore Website (the awards syllabus, The Singapore Life Saving Society, pages 24 to 31) for the types of Open Water Lifesaving certifications.

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NOTE 2 – Refer to the Sport Singapore website on the national registry of coaches (NROC) for more information on the criteria to be a NROC coach.

4.5.3 Where prior experience may be recognised, a portfolio of sufficient training (at least 80 h, or specific to the activity) and activity delivery (300 h, or specific to the activity) should be available for verification.

4.5.4 Coaches and trainers shall ensure the following:

a) Understand the local conditions, including weather, tides, currents, shipping/vessel traffic, physical obstructions, shelters, and other potential hazards. They shall perform pre-activity/event risk assessments, raise awareness of the restrictions and possible risks of the open water activities and facilities, and monitor and guide participants on safe use or areas to avoid;

b) Routinely check the weather forecast and tide conditions for the duration of the activity;

NOTE – Refer to the Meteorological Service Singapore website for weather forecasts, warnings and advisories.

c) Understand that all participants have different capabilities and can react differently to the training provided. They shall have knowledge of participants' health status and limitations (e.g. through participants' declaration forms and pre-participation screening prior to activities);

d) Plan for a reasonable coach/trainer-participant ratio (e.g. the recommended ratio for improvised rafting is 1:14);

e) Conduct open water orientation and safety training on relevant types of safety equipment and what to do during distress. Key pre-activity safety knowledge shall cover the following:

- How to wear the PFD and its use for self-recovery;
- What to do in the event of a capsized craft (capsize drill);
- How to use the equipment provided (skills orientation/training);
- What to do during inclement weather; and
- What to do in case of emergency when out in the open water (emergency plans).

f) Supervise the participants and trainees closely and keep them within the line of sight. They shall group and conduct the activity or training based on the participants' capabilities. They shall limit the area of open water activities to maintain supervision and control. Where possible, coaches and trainers shall maintain close proximity with participants and trainees to allow for quick access and response in the event that any of them face difficulty in the water;

g) Routinely reassess risks (to people and environment) during the activities and promptly intervene to mitigate them when necessary;

h) Be sufficiently equipped to communicate (e.g. with whistles and communication devices such as 2-way radios) with other coaches, trainers, conducting personnel, activity instructors and guides and respond to a hazard or an emergency. They shall also have immediate access to rescue and safety equipment (e.g. PFDs, rescue tubes, towlines, throw bags, light sources, emergency blankets, etc.);

i) Be familiar with the emergency response plan in case of incidents. They shall perform the roles of a lifeguard in the absence of designated lifeguard;

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- j) Allow frequent breaks during activities for hydration and rest;
- k) Take direct responsibility for participants' safety and tailor programmes to suit their skills and needs;
- l) Report major or drowning incidents to facilities owners/occupiers and organisers; and
- m) Conduct post-training evaluations, submit reports to their respective agencies, and maintain these reports for easy access.

4.5.5 Coaches and trainers of activities should use the checklist for safe water activities in Annex B before the commencement of the activities.

4.6 Parent/adult guardian/caregiver

4.6.1 Constant and close supervision by this group of stakeholders can prevent drowning incidents. The availability of trained lifeguards and first aiders, complemented by the close supervision of parents, adult guardians and caregivers, can reduce the risk of drowning complications and incidents. Parents, adult guardians and caregivers should have basic lifesaving and swimming skills when bringing children who are non-swimmers or weak swimmers to open water locations or facilities.

4.6.2 Parents, adult guardians and caregivers shall be responsible for their children, especially those under 15 years old, through some level of supervision such as:

- a) being in the water, within arm's reach and actively supervising children aged 0 to 4 years;
- b) actively supervising in the water and/or from the water's edge for those aged 5 to 10 years; and
- c) regularly checking on those aged 11 to 14 years.

4.6.3 Parents, adult guardians and caregivers shall ensure the following:

- a) Be cognisant of their child's capabilities and only allow them to swim in open water if they already know how to swim but always under close, active supervision. If the parent, adult guardian or caregiver does not know how to swim, the child shall stay at the beach area and away from any water activities.
- b) Be aware of the child's health (including pre-existing conditions) and physical well-being. Thereafter, have all necessary precautionary measures in place before allowing the child to participate in any open water activities.
- c) Equip the child with PFDs and ensure that they are worn throughout the water activity.
- d) Be familiar with the open water conditions, whether local or overseas. Gather relevant information of the open water facility to determine if it is safe before allowing the child to attempt open water activities (e.g. weather, tide, water conditions, possibility of aquatic animal stings or bites) or other incompatible activities within the area, such as swimming, kayaking etc.
- e) Be aware of and familiar with rescue protocols (refer to Figure 1) and emergency contacts in case of any predicament.

NOTE – Refer to the medical position statement – MPS 19, Drowning chain of survival, International Life Saving Federation.

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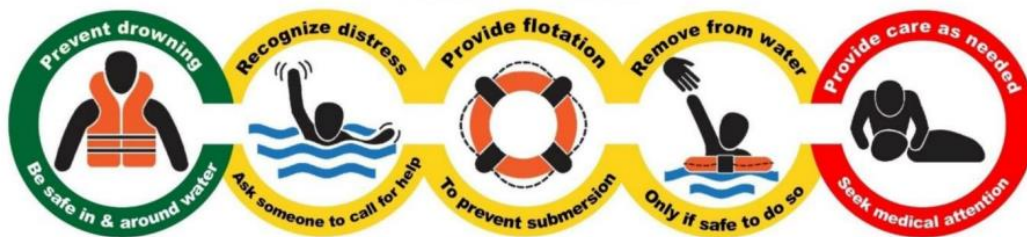


Figure 1 – Drowning chain of survival

4.6.4 Parents, adult guardians or caregivers should learn first aid including CPR and AED skills to perform the roles of first responder prior to arrival of an ambulance and medically trained personnel.

4.6.5 Parents, adult guardians or caregivers shall abide by the following safety regulations, especially those displayed at open water facilities:

- a) Always swim between the red and yellow flags;
- b) Do not swim after consumption of alcohol or any medication that can cause drowsiness or impair judgement before participation;
- c) Always swim with a buddy and stay close to the shore;
- d) Always actively supervise children at the beach;
- e) Check the weather and water conditions before swimming and never swim at night, when there is low visibility (e.g. heavy fog);
- f) Reduce exposure to the sun (e.g. apply sunblock, wear proper clothing, and drink plenty of water);
- g) Stay away from rocks and other hazards;
- h) Use emergency intercoms for emergencies, if available at first-aid stations; and
- i) Pay attention to safety advice by beach patrol officers.

4.6.6 Parents, adult guardians or caregivers should seek professional assistance to provide open water orientation, if they lack experience or competence in such activities, for their children.

5 Buddy system

5.1 Looking after each other's safety

The buddy system is an important risk control measure where participants are paired or teamed up to look after each other's safety. When used during open water activities such as water safety orientation, this system can be the first line of defence for risks (e.g. drowning, entrapment and collision) by either providing immediate assistance or getting the attention of safety personnel. Participants share the responsibility for their buddy's well-being and safety. This type of support is important in any activity. Examples of buddy-up include:

- a) participants who are experienced and less experienced;
- b) swimmers and non-swimmers; and
- c) medically cleared and those with medical issues.

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5.2 Deployment of the buddy system

Coaches, trainers, conducting personnel, activity instructors and guides should brief participants on the following prior to deploying the buddy system:

- a) Deploy a two-person team (minimum).
Understand the buddy's background, prior experience, and job role (if different from theirs). Stay close to the buddy and communicate regularly.
- b) Look out for hazardous conditions, safety demands and stressors.
Identify the buddy's strengths and weaknesses. Monitor the buddy for stress (including environmental stress). Check on the buddy and encourage regular breaks. Observe one another when putting on and taking off personal protective equipment (PPE).
- c) Manage stress to prevent burnout.
Communicate and make their buddy feel safe to speak out. Encourage routine sleep, exercise and eating. Acknowledge tough situations, recognise accomplishments/burnout, and discuss their impact. Identify opportunities for stress relief. Talk with one another about their experience.

6 Enhancing supervision with technology

Supervision should be improved with the use of technology. As technology is constantly evolving, the list below is not exhaustive but should be considered for use by all stakeholders:

- a) Two-way radios
Supervision should encompass two-way radios for improved communication, especially in scenarios where line-of-sight communication is not feasible. Two-way radios also facilitate coordination amongst safety personnel, ensuring comprehensive supervision and rapid response, especially when the event locations are a distance away from shore.
- b) Mobile and portable phones
These provide quick contact with emergency services. Supervisory staff should have their mobile phones on hand to minimise any delay in reporting incidents. Staff on duty shall not use their mobile phones or portable phones for personal reasons.
- c) Security alarms
Areas that are deemed dangerous or not open to the public should be installed with security alarms for notifications of breach of security to the closed area.
- d) Emergency alarms/announcements
These should be provided at open water events, locations, and facilities for any of the following situations:
 - Inclement weather and lightning alerts;
 - Change of water flow conditions and opening of tidal gates at reservoirs; and
 - Dangerous occurrence and incompatible open water activities spotted.
- e) Closed circuit televisions (CCTVs)

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Manned CCTVs, where applicable, should be installed at locations where there are many participants as they provide “bird’s-eye view” of the surroundings. Lifeguards or security personnel on the ground should be notified immediately of persons in distress using two-way communication walkie talkies. Deployment of CCTVs should include the following conditions:

- Install CCTVs at elevated positions for broader coverage;
- Ensure no physical obstructions of the CCTVs (e.g. trees);
- Maintain CCTVs well for continuous monitoring;
- Deploy additional CCTVs particularly in high-risk areas; and
- Be sensitive to privacy considerations when installing CCTVs.

f) Drones

These are commonly used to reach areas where it is not feasible to conduct supervision. Drones should be considered as they provide a bird's-eye view of open water activities. For example, the clear footages provided of incidents enable organisers to identify the impact of injury before persons in distress reach the shore. Drones should be used for events or activities if there are limited numbers of safety boats. It is presupposed that all unmanned aircraft operations are conducted in compliance with applicable statutory and regulatory requirements.

7 Drugs and alcohol

7.1 Consumption of alcohol or recreational drugs in, on, or around water can increase the risks of drowning by impairing alertness, judgement, coordination and reaction time.

7.2 Participants who have consumed alcohol or drugs shall abstain from open water activities.

7.3 Participants taking prescribed medication shall do the following:

- a) Consult doctors for safety advice before participating in open water activities;
- b) Take any prescribed medication as directed by doctors;
- c) Be aware of any medication side effects; and
- d) Understand that medication can affect their ability to supervise others near water.

7.4 For open water events, alcohol provision and open water activities shall be segregated by deploying physical barriers. Warning signs shall also be displayed to remind participants of the risks of drowning.

7.5 Organisers shall conduct pre-event briefings to remind relevant stakeholders about the dangers of participating in water activities whilst under medication or under the influence of alcohol. Anyone appearing drowsy or disorientated shall be removed from the water activities.

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8 Safety management

8.1 Events/competitions

8.1.1 Event organisers shall prepare and include an event safety plan as part of the overall event management plan.

8.1.2 The event safety plan shall include the following key provisions:

- a) Medical – Medical and rescue coverage shall be provided for all open water events and competitions:
- b) Deployment of lifeguards and/or safety boats – The maximum ratio of assigned lifeguards to participants shall not be more than 1:20.
- c) Number of officials, marshals and boats – They shall be determined by the:
 - type and nature of events;
 - number of participants;
 - experience and age of the participants; and
 - location, proximity and number of water egress points.
- d) Communication – The event organiser shall establish the communications requirements, include them in the event safety plan and ensure proper equipment use.
- e) Event information – Participants shall be informed of the water conditions, emergency action plan, course/activity areas, safety provisions and health and safety related information before the event.
- f) Safety briefing – Participants shall be briefed on any safety concerns and instructions before the event;
- g) Medical coverage – First-aid cover, such as trained first aiders and equipment, shall be provided. Consideration should be given to the deployment of trained first aiders, placement of lifesaving and rescue equipment, and whether there is a need for first-aid expertise on safety boats.
- h) Rescue equipment – Adequate rescue equipment such as throwlines, life buoys, rescue tubes and any other floatation devices shall be made readily available for use in response to any incidents.
- i) Emergency procedures – The triggers for shortening the course or abandoning the event shall be included in the procedures. These triggers include:
 - maximum safe sea state;
 - maximum wind speed;
 - poor visibility;
 - forecast of an imminent thunderstorm; and
 - interference by other boat participants.

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- j) Emergency response and management plan – Procedures for activating advanced medical care shall be planned. This includes the location of first-aid stations, the nearest hospital, and ambulance access and evacuation routes. Event organisers shall also conduct drills/tests for water rescue, transfer to rescue vehicles, moving to shore, transfer from shore to ambulance, and also a practice run of the route for the ambulance or admin vehicle, and make any improvements relating to equipment, staffing skills, environment route, etc. where necessary.

8.2 Risk management

8.2.1 Risk management, the process of identifying, evaluating and controlling risks at the workplace, is essential as it forms an integral part of a safety management system. Risk management involves:

- a) conducting risk assessments of work activities;
- b) controlling and monitoring the risks of work activities; and
- c) communicating the risks to the relevant stakeholders.

8.2.2 The responsibility for conducting risk assessments shall lie with the stakeholders, although a designated third party such as a safety officer can be engaged to document the risk assessment and management processes. Key factors to include in the risk assessment plan are equipment, physical, environment, processes, and hazards relating to participants (refer to the list of examples in Annex C).

8.2.3 Documenting risk assessment findings helps track risks and control measures, identifying issues before they become problems. Safety policies, procedures, inspections, etc. shall be documented for a safety audit trail. These records demonstrate a responsible and proactive approach to managing safety and provide templates for future events. The risk assessment and associated safety documents shall be periodically reviewed to remain relevant. For example, in triathlons, water temperature shall be measured 1 h prior to the start of the event. It shall be taken at the middle of the course and in two other areas on the swim course, measured at a depth of 60 cm. Where the average measured figure is 27 °C or below, the lowest measured temperature can be considered as the official water temperature. Where the average measured figure is above 27 °C, the highest measured temperature can be considered as the official water temperature.

8.2.4 Relevant stakeholders shall carry out risk management, in areas such as (but are limited to) the following:

- a) Conduct risk assessments and any subsequent updates at the locations/facilities;
- b) Review risk assessment plans once every three years or whenever there are any significant deviations from approved processes or designs, introduction of new equipment, upgrade of facilities, procedures or occurrence of injuries or incidents;
- c) Validate that safety measures (identified within the documents) are in place and any deviations on the event day shall be documented prior to the commencement of the event; and
- d) Implement a risk management plan and supervise programmes effectively.

NOTE – Refer to the Workplace Safety and Health Council website for the Code of practice on workplace safety and health (WSH) risk management.

8.2.5 If risks cannot be managed to acceptable levels, steps shall be taken to completely avoid the hazard (e.g. by changing the start time of the event). If this is not possible, then the activity shall not take place. For more information on risk assessment, refer to Tables D.1 to D.5 in Annex D.

NOTE – Refer to the Workplace Safety and Health Council website for a sample of the risk assessment form.

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8.3 Standard operating procedure (SOP)

8.3.1 Relevant stakeholders shall refer to the risk assessment plan to draw up SOPs from pre-event checks to post-event debrief.

8.3.2 For training sessions, organisers, coaches or trainers, and conducting personnel, activity instructors or guides should draw up specific rules relating to their activity for briefing to users. A copy should also be extended to the participants and for them to acknowledge and indicate that they have read and understood the information.

NOTE – Refer to the Sport Singapore website for the simple guide to risk assessment management.

9 Personal responsibility

9.1 Pre-participation screening

Participants shall take the necessary preventive measures for their own health and safety during open water activities, including completing pre-participation screening. In case of non-conformity in the pre-participation screening, they shall refrain from participation in open water activities until consulting a doctor and/or additional safety measures are put in place.

9.2 Personal flotation devices (PFDs)

9.2.1 Participants in open water activities shall be responsible for their own safety and use appropriate PFDs such as life vests (inflatable or non-inflatable), inflated floats and swimming floats. The most common PFDs are buoyancy aids and life jackets. Life jackets, suitable for all ages and swimming abilities, reduce drowning risk and should be used by non-swimmers and weak swimmers in all open water activities. Participants shall not rely on air-filled or foam toys as safety devices. Life jackets when worn shall be well-fitted and comfortable. Children under 13 years of age should have appropriate sizes and types of life jackets. Participants shall conduct a simple fit test by lifting the jacket straight up from their shoulders and ensure it does not ride up. PFDs used should conform to International Standards such as ISO 12402-4. However, in open waters where the sea conditions can be more challenging, a higher performance level like 150 or 275 should then be applied (refer to the ISO 12402 series).

9.2.2 Participants shall be responsible for physical checks, maintenance, and replacement of their personal PFDs. Stakeholders supplying PFDs for use by participants shall provide fully functional equipment and shall be responsible for their periodic checks, maintenance, and replacement of the PFDs. Selection of the appropriate type of floatation devices depends on the water conditions and the time expected for the rescuers to arrive. The different types of PFDs are described as follows and illustrated in Figure 2:

- a) Type I – Offshore life jackets (minimum buoyancy of 10 kg for adults and 5 kg for children). These provide the best protection and are best for open, rough, or remote waters where rescue can be delayed, and are useful for cruising, racing, fishing offshore, boating alone or in stormy conditions. These lifejackets can turn most unconscious wearers face-up in water.
- b) Type II – Near-shore buoyancy vests (buoyancy ranges from 7 to 10 kg). These are suitable for calm, protected, and inland waters where chance of quick rescue is good. These life jackets can turn some unconscious wearers face-up in water. They are not suitable for extended survival in rough water and often require users to tread water to keep the head above water. Their intended use is for inland day cruising, fishing, sailing, and boating in light craft.
- c) Type III – Floatation aids (minimum buoyancy of 7 kg). These are comfortable and suitable for users in calm or inland waters and intended for use at supervised activities such as sailing

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regattas, dinghy races, water skiing, fishing, canoeing, kayaking, and during personal watercraft operation. To remain in a face-up position in water, users may have to tilt their head back.

- d) Type IV – Throwable devices (buoyancy ranges from 7.5 to 8 kg). These are not to be worn and designed to be thrown to an overboard victim or to supplement the buoyancy of a person overboard. These devices are not meant for unconscious persons, non-swimmers, or children. They should be located within an arm's length so that it can be thrown over the side immediately in an emergency.
- e) Type V – Special-use devices (buoyancy ranges from 7 to 10 kg). These are optimised for activities such as kayaking and sailing. Many paddlers wear a Type V vest with a quick-release tab attached to a line and ring meant for live-bait rescue. Prior to wearing this vest, participants should receive proper training in its various functionalities.

NOTE – Although Types I and II are the commonly used PFDs, the other types of PFDs work as well. Refer to the Coastal Boating website for other types of PFDs.



Figure 2 – Types of PFDs

9.3 Acclimatisation

Participants of open water locations, facilities, and activities shall acclimatise themselves or be accustomed to changes in climate or to new conditions (e.g. water can be cold and a drop in body temperature can cause hypothermia). They should have previously experienced or undergone open water familiarisation courses or have competent persons to guide and accompany them. Participants should adopt the following measures (non-exhaustive):

- a) Acclimatise to water temperature;
- b) Avoid entering open waters at water temperature below 10 °C;
- c) Shorten the exposure when water temperature is below 14 °C;
- d) Use wet or dry suits to provide thermal protection;
- e) Anticipate and prepare for the weather and water conditions (e.g. changes in depth, temperature, currents and turbulence that can occur without warning); and
- f) Be aware of one's fitness level and avoid venturing too far out from shore.

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9.4 Ultraviolet (UV) protection

Open water activities have a risk of over exposure to UV rays, which can then cause acute and chronic harmful effects on the skin, eyes, and immune system. Participants should adopt the following general sun protection tips:

- a) Plan outdoor activities to avoid the sun's strongest rays (generally from 11 am to 3 pm).
- b) Minimise duration in the water, on the water, and on sand to reduce chance of sunburn and damage to skin and eyes.
- c) Wear protective clothing such as wide-brimmed hats, long pants and long-sleeved shirts made of tightly woven fabric to reduce sun exposure.
- d) Wear sunglasses that provide 100 % UV ray protection (both UV-A and UV-B). Large, wrap-around style frames may provide more efficient UV protection because they cover the entire eye-socket. This is especially important when doing activities around, in, or on water because much of the UV comes from light reflected off the water.
- e) Use reef-safe sunscreen and/or a broad spectrum (protecting from both UV-A and UV-B) sunscreen with a sun protection factor (SPF) and reapply every 1.5 to 2 h even on cloudy days and after swimming or sweating. Both selection of sunscreen and re-applications are important. Carefully examine your skin once a month as early detection of melanoma can save one's life. Consult with a dermatologist if there are unusual moles, new or changing skin lesions, scaly patches or sores with local persistent bleeding or that does not heal.
- f) Use wide-brimmed hats (brims of at least 10 cm wide) that are made of tightly woven, opaque fibres that shield one's face and shoulders for the most protection.
- g) Wear light-coloured and lightweight fabrics for the most comfort and protection such as sun-protective swimsuits.

10 Swimming caps

10.1 Swimming caps can help in segregating participants by proficiency. Organisers for open water activities should:

- a) issue brightly coloured swimming caps to participants; and
- b) group swimmers with similar proficiency using same coloured swimming caps for better management and avoid collision amongst swimmers.

10.2 Participants at open water locations should use snug-fitting, conspicuous swimming caps.

11 Open water safety equipment for underwater sports

11.1 Dive flag

Underwater sports, like scuba diving, involve high risks. Therefore, scuba divers shall make use of safety equipment including dive flags, surface marker buoys and PPE. A dive flag shall be hoisted to signal "stay clear, divers below" to boaters.

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11.2 Surface marker buoy (SMB)

Dive leaders shall deploy SMBs at the end of their dive to indicate their position and warn boaters to keep clear and stay at a distance during their decompression stop underwater. SMBs shall also be used when free divers are involved in apnoea training (e.g. conditioning the body to stay underwater without air and oxygen) in open water conditions.

11.3 Underwater diving equipment

The following equipment should be used for underwater diving (non-exhaustive), where appropriate:

- a) SMBs, dive knife/cutter, duck horn/tongs, torchlight and whistle;
- b) Wetsuit, hood, glove, bootie, open-heel fins, mask and snorkel;
- c) Scuba tank, regulator, buoyancy compensator, weights and belt; and
- d) First-aid box and oxygen cylinder.

12 Locations/facilities and environment

12.1 No-swim zones

12.1.1 Stakeholders of open water locations or facilities shall assess and determine the environmental risks, climate changes and presence of dangerous sea creatures before deciding to close the location/facility if necessary. Any closure shall be made known to the public via information boards, flags, website, etc.

12.1.2 The following are the key conditions for determining no-swim zones:

- a) Contaminated water posing significant health hazards;
- b) Persistent outward water currents, which make swimming back to shore difficult;
- c) Strong water currents or turbulence;
- d) Incompatible water activities such as canoeing, sailing, rafting, etc.;
- e) Presence of hostile sea creatures (e.g. jellyfish); and
- f) Risk of lightning or strong water turbulence due to climate changes.

12.1.3 Participants of open water activities shall respect the decision of the stakeholders in the event of any closure or restriction of access to locations or facilities.

12.2 Buoys for demarcation

12.2.1 Buoys used for demarcation to segregate the different purposes of the open water locations/facilities shall comply with the following:

- a) Brightly coloured and easily visible from afar;
- b) Securely fastened and be able to withstand water turbulence; and
- c) Appropriately spaced out.

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12.2.2 Relevant stakeholders shall maintain, repair, and replace demarcation buoys to ensure they remain in good serviceable condition.

12.3 Protection against lightning

Owners/Occupiers of open water locations or facilities should refer to the SS 555 series.

12.4 Signage and information boards

12.4.1 Owners/occupiers of open water locations or facilities shall determine critical information for communication to the relevant stakeholders and the public, such as no-swim zones, flag representation, depth of open water, restricted water activities and sudden drop of water. Information boards shall have the following requirements:

- a) Easily comprehensible current and relevant information (including complementing pictures or graphics) for stakeholders, including the public, regardless of age or nationalities; and
- b) Easily accessible, conspicuously coloured, and easily legible.

12.4.2 Owners/occupiers shall maintain, repair and replace the information boards to ensure they remain in legible condition.

12.5 Protection from over exposure to heat

12.5.1 Hot and humid weather conditions can cause participants to be over-exposed to heat with risks increasing when they are performing high-intensity activities. The body typically cools itself by sweating. During hot weather, especially with high humidity, body temperature can rise to dangerous levels if there is insufficient hydration and rest leading to heat exhaustion or heat stroke.

12.5.2 Heat injury can be prevented by using the following methods:

- a) Acclimatise to local conditions;
- b) Avoid strenuous activities during hot weather;
- c) Reduce activity duration;
- d) Reduce activity intensity;
- e) Provide shelter where possible;
- f) Provide ice, ice baths, and other cooling devices where appropriate;
- g) Increase hydration at frequent intervals; and
- h) Wear protective clothing.

12.5.3 It is important to recognise the symptoms of heat injuries, which include the following:

- a) Symptoms of heat injuries (as experienced by athletes):
 - Giddiness and dizziness;
 - Shortness of breath even at rest;
 - Numbness, cramps, weakness of extremities;

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- Light-headedness, fatigue, drowsiness;
 - Nausea; and
 - Headache.
- b) Signs of heat injuries (as witnessed by onlookers):
- Dry and warm skin;
 - Altered mental status (confusion, disorientation, combativeness);
 - Ataxia (inability to maintain balance, frequent falling);
 - Hyperventilation;
 - Vomiting;
 - Loss of consciousness; and
 - Seizures.

12.5.4 When someone is identified as suffering from heat injury, dial the emergency hotline (i.e. in Singapore, 995) or, if there is onsite medical coverage, alert the medical post so they can respond.

While the first responders attend to the emergency, the following actions shall be taken:

- a) Conduct rapid cooling;
- b) Bring the person to a cool, shaded area;
- c) Provide water for drinking;
- d) Sponge the person;
- e) Assess the person's level of consciousness; and
- f) Promptly seek further medical treatment, especially if the person is confused, disorientated, experiencing rapid and shallow breathing or unconscious.

12.6 Protection from marine animals

12.6.1 Open water is home to a wide variety of marine life, and not all are friendly. Participants should take the following precautions to avoid being stung or ensure easy access to medical aid:

- a) Always swim near lifeguards so that immediate first aid or, in severe cases, a means of contacting an ambulance is readily available.
- b) Exercise caution when walking in rocky areas or near seaweed, as sea urchins are often found in shallow, rocky areas.
- c) Avoid touching or handling stinging sea creatures, as their tentacles can cause stings even when they are dead.
- d) Pay attention to warning signs on the beach. These signs are often put up after sightings of groups of jellyfishes or stonefish.

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- e) Scuff their feet when walking in shallow water to warn and scare off any stinging sea creatures in their path.
- f) Wear protective clothing, such as wetsuits or protective footwear, as stonefish, razor clams, stingrays and sea urchins are often found in shallow water. Rubber-soled shoes or sandals can help protect one's feet while in shallow water or rocky areas. Wearing a wetsuit can also provide protection against jellyfish stings during prolonged swimming.

12.6.2 Participants should adhere to the following list of safety recommendations:

- a) Engage in sea sports with a buddy who is able to look out for them;
- b) Minimise exposed skin by wearing a long-sleeved shirt/rash guard and pants, or a wetsuit;
- c) Use a flotation device such as a safety buoy or life jacket;
- d) Have a bottle of plain vinegar readily available; and

NOTE – Plain vinegar reduces the effect of jellyfish stings.

- e) Refrain from approaching or touching any jellyfish, whether in the water or on the beach.

NOTE – Refer to the National Parks Board website for animal advisories on box jellyfish.

13 Water quality

13.1 General

Water activities inevitably expose users to water and, in some instances, lead to unintentional water consumption.

13.2 General precautions for beach users

Beach users should take the following precautions:

- a) Avoid water-based recreational activities if they have open sores, skin infections or are unwell.
- b) Avoid touching their mouth and/or eyes with hands that have been in contact with sand and/or water.
- c) Practise good personal hygiene such as rinsing after swimming and washing hands thoroughly before eating or handling food.
- d) Avoid areas with visible algae concentration or scum in the water as well as on the shore. Where no scum is visible but the water shows a strong greenish discolouration and turbidity, beach users should ensure they can still see their feet when standing knee-deep in the water. If not, avoid bathing or ingesting the water. In addition, avoid water skiing because of potentially substantial exposure to aerosol.
- e) If sailing, sailboarding, or undertaking any activity likely to involve accidental water immersion in the presence of cyanobacterial or algae blooms, wear snug-fitting clothing to minimise exposure. Using wetsuits for water sports can increase the risk of rashes because cyanobacterial or algae material in the water trapped inside the wetsuit can remain in contact with the skin for an extended period.

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13.3 General precautions for organisers

Infections and illness resulting from recreational water activities are generally mild and difficult to detect through routine surveillance systems. Even when illnesses are more severe, attributing them to water exposure can be difficult. Organisers shall ensure the following:

- a) Conduct risk management to address possible infections and illnesses during and after activities;
- b) Raise awareness among participants about potential risks and general precautionary measures, such as those listed above;
- c) Select appropriate locations for water activities, focusing on sites rated “Good” or “Very Good”;
- d) Test water quality using accredited testing authorities;
- e) Conduct visual inspections of potential sources of contamination;
- f) Ensure that there are sufficient facilities for the number of participants (e.g. number of participants to bathing facilities and beach);
- g) Provide immediate first aid/urgent medical transport; and
- h) Encourage participants to declare their health and medical history.

13.4 General precautions for owners/occupiers

Owners/occupiers are unlikely to provide supervision for recreational water especially if it is open to the public year-round. However, they shall implement the following safety measures:

- a) Conduct regular monitoring and auditing through minor and major incident reports;
- b) Provide public information about the recreational water environment through websites/advisories, including updates on improvements in conditions;
- c) Develop long-term measures like pollution abatement and prevention; and
- d) In cases of poor water quality, issue notices/circulars/advisories including but not limited to signposting, information via website and posters to discourage its use.

14 Seasickness

14.1 Seasickness while out in open water can cause nausea, fainting and vomiting. Prolonged seasickness can lead to dehydration and, if left untreated, can even lead to heat injury (e.g. heat exhaustion).

14.2 Organisers/operators should practise the following:

- a) Identify the signs and symptoms of seasickness and provide first aid or mitigation measures;
- b) Check if participants have a known history of motion sickness;
- c) Ensure participants are well-rested and hydrated before any activity;
- d) Ensure participants carry sufficient accessible drinking water for the duration of the activity;

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- e) Monitor and supervise participants' periodic intake of water during the activity; and
- f) Remove participants from open water environment if the condition does not improve after first aid or mitigation measures.

14.3 To prevent seasickness, participants can try the following tips:

- a) Keep their eyes on the horizon to help their body adjust to the motion of the ship/boat;
- b) Stay well-hydrated and avoid heavy, greasy or spicy foods.
- c) Choose a cabin in the middle of the ship where the motion is less noticeable (if applicable) and avoid closed cabins; and
- d) Consume seasickness medication before the activity.

15 Outreach and education – Water safety

15.1.1 Many participants do not realise the dangers of swimming or participation in open water activities. Water safety education and understanding about currents, drop offs, water visibility, undertows, waves, and underwater obstacles, including seeking help when in distress, can help save lives.

15.1.2 Safety information about the locations, facilities or events shall be made available to participants prior to event and/or at the site. Key safety information on safe use of the locations or facilities shall include the following:

- a) A general map of the facilities or event site;
- b) Dangers or out-of-bounds areas;

NOTE – Out-of-bounds areas are those permanently disallowed access.
- c) Off-limits areas and no-activity hours;

NOTE – Off-limits areas are those that are temporarily blocked from entry.
- d) Procedures to follow in inclement weather;
- e) Basic first aid and emergency procedures, including emergency hotline numbers and guidance for dealing with stings, cuts, heat injuries and drowning; and
- f) Exclusions, specifying conditions under which users should not enter the water (e.g. when alone, intoxicated, medicated, bad weather, strong tidal currents, etc.).

16 Emergency preparedness

16.1 Emergency response plan (ERP)

16.1.1 Relevant stakeholders shall be prepared for any unforeseen situations and have the necessary resources to carry out rescue operations. They shall develop an ERP to better equip and prepare for unexpected events. These plans shall be based on the worst-case scenarios and immediate responses to alleviate injuries or prolong life. Situations include, but are not limited to, injuries to swimmers, hypothermia, drowning, distressed swimmers, inclement weather such as lightning, high winds, poor visibility, and unauthorised crafts entering the swimming zone.

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16.1.2 The ERP shall include the following key information and terms:

- a) Roles and responsibilities;
- b) Communication (on-site and externally, e.g. ambulance services);
- c) Provision of medical coverage and necessary equipment such as blankets;
- d) Access to safety equipment (e.g. throw ropes, life buoys, emergency vehicles, etc.);
- e) Routes of evacuation; and
- f) Rescue and recovery of swimmers from the water.

16.1.3 Practice scenarios, drills, and tabletop exercises shall be conducted to train assigned roles and ensure everyone understands what to do in an emergency. Such exercises also test the effectiveness of the ERP.

16.1.4 A systematic approach to identify potential hazards, such as the 5M model (man, machine, medium, management and mission for risk management), shall be used when deriving the ERP and countermeasures for prevention and incident management.

16.1.5 Reviews shall be conducted to ensure the relevance of emergency plans for open water locations, facilities, events, and activities. Changes to the activities or the introduction of new equipment shall involve a review and any necessary adjustments to the emergency plans.

16.2 Emergency responder

There shall be designated emergency responders, especially for organised activities where large crowds are expected and the risk of an incident occurring is higher. The deployment of emergency responders shall be determined by relevant stakeholders in their ERP. These stakeholders need to assess the likelihood of incidents to deploy the appropriate numbers of trained emergency responders.

16.3 Emergency equipment

Relevant stakeholders shall determine the necessary emergency equipment, which includes the following (non-exhaustive):

- a) AEDs;
- b) Ambulance;
- c) First-aid boxes;
- d) Loudhailers;
- e) Mobile phones;
- f) Rescue tubes;
- g) Safety boats;
- h) Stretchers/cover litters;
- i) Two-way radios; and
- j) Washing/bathing facilities.

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16.4 Emergency routes

Open water locations/facilities shall have designated emergency vehicle parking lots and accessible routes. Relevant stakeholders shall designate emergency routes where practicable. Emergency routes shall be:

- a) as close to open water locations/facilities as practicably possible;
- b) shown with parking lots with sufficient space for emergency vehicles to enter and exit easily;
- c) restricted to the public and controlled with physical barriers and signages; and
- d) indicated on location maps to direct emergency vehicles.

16.5 First-aid rooms

16.5.1 Open water locations/facilities that have high patronage or hold frequent activities shall be equipped with first-aid rooms and trained personnel. First-aid rooms or areas shall be dedicated to the provision of first aid only and should be adequately cleaned and disinfected after each use. However, if a multi-purpose area is chosen to be a first-aid room, the first-aid facilities and equipment shall always be clear of obstruction, easily accessible, and ready for immediate use. The first-aid room shall be equipped with the following (non-exhaustive):

- a) AED;
- b) Chairs;
- c) Container for collection and disposal of soiled medical items used in first aid;
- d) First-aid boxes;
- e) General power outlets;
- f) Paper towel and dispenser;
- g) Portable oxygen equipment;
- h) Removable screen to protect privacy of casualty;
- i) Safety data sheet (where applicable);
- j) Sink;
- k) Soap and nail brush;
- l) Stretchers;
- m) Sufficient lighting;
- n) Torches (pocket size); and
- o) Work bench or dressing trolley.

16.5.2 Relevant stakeholders shall maintain and replace the equipment in the room to ensure that there is an adequate quantity of items and that they are in working condition (i.e. ready to be used).

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16.6 First-aid treatment

First-aid treatment shall be provided by trained personnel. They shall be familiar with the ERP, layout, and equipping of the first-aid room.

16.7 Safety support (safety boat)

16.7.1 Activities, such as canoeing and sailing, that involve groups of participants shall have safety boats deployed and readily available on the water to respond to emergencies. It is presupposed that lifesaving equipment on safety boats and/or pleasure crafts are in compliance with applicable statutory and regulatory requirements. Lifesavers on rescue boats shall wear PFDs at all times, hold valid First Aid and AED certificates, and trained in basic lifesaving skills and basic swimming skills.

16.7.2 Relevant stakeholders shall service, maintain and replace the safety boats regularly to ensure they are in good and serviceable condition (refer to boat checklist in Annex E).

16.8 Rescue and lifesaving equipment

16.8.1 Rescue and lifesaving equipment comprise lifebuoys and rescue tubes, AEDs/first aid boxes, stretchers/cover litters.

16.8.2 Lifebuoys and rescue tubes are essential for conducting rescue operations in the event of an incident. Relevant stakeholders shall determine the number of lifebuoys and rescue tubes based on their respective locations and facilities. They shall service, maintain and replace these items regularly to ensure they are in good and serviceable condition. Lifebuoys and rescue tubes shall meet the following requirements:

- a) Installed at premises where there is a risk of drowning;
- b) Readily accessible and not locked;
- c) Conspicuous and brightly coloured;
- d) Neither too heavy to handle nor too light to throw out, and
- e) Provide sufficient buoyancy.

16.8.3 AEDs, first-aid boxes, stretchers, and cover litters are critical use during emergencies. Relevant stakeholders shall determine the practicable quantity of the equipment based on the average number of users. The equipment shall be readily available, easily identifiable with clear signage and guidance on its use. The equipment shall be maintained, serviced, and replaced to ensure there is an adequate quantity of functional equipment (i.e. not expired).

16.8.4 The contents of the first-aid box should include the following items in reasonable quantities to cater to expected injuries resulting from activities conducted:

- a) Adhesive dressing (plasters);
- b) Adhesive tape (hypoallergenic);
- c) Antiseptic solution/cream;
- d) Burns dressing;
- e) Disposable gloves;
- f) Gauze pieces;

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- g) Safety pins;
- h) Scissors;
- i) Thermometer;
- j) Triangular and crepe bandages;
- k) Tweezers; and
- l) Wound dressing.

16.9 Medical team and safety vehicle (ambulance)

The need for on-site medical cover (e.g. by an emergency medical technician or medical doctor) shall be determined based on the nature and size of the event. Organisers of events shall ensure that ambulances are on standby when necessary. Relevant stakeholders shall comply with Clause 10 of SS 681:2022.

NOTE – Refer to the Ministry of Health website for the types of qualifications for medical personnel, such as paramedics.

17 Incident reporting and management

17.1 Investigation and reports

17.1.1 Relevant stakeholders shall prepare incident reports for all incidents, especially major incidents, drownings, and near-drowning cases. This data collection can provide vital statistics and information for trend analysis, continuous improvements, and prevention of similar occurrences.

17.1.2 The primary purpose of a thorough incident investigation is to determine the root cause so that appropriate corrective measures can be implemented. The incident investigation shall be conducted as soon as practically possible. The timeline of events leading to incident and those following it shall be recorded. Photographs and other evidence shall be identified as early as possible. Key witnesses shall be interviewed to gather as much information as possible. Incident reports shall include, but not be limited to, the following:

- a) Location;
- b) Date/time of incident;
- c) Activity of casualty;
- d) Casualty particulars (e.g. identity and contacts);
- e) Type of injuries;
- f) First aid treatment provided;
- g) Any AED shocks administered;
- h) Hospital/medical centre conveyed to;
- i) Witness (if any) and witness statements;
- j) Root cause (if any); and
- k) Corrective actions (if any).

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17.1.3 Incident reports serve as valuable lessons learnt, and there shall be proper documentation and record-keeping. Incident reports shall be retained for a period of at least three years. The follow-up actions, including the review of corrective measure, shall be documented and submitted to their respective management. These documents should also be submitted to a central and independent body.

18 Preventing collisions at sea

It is presupposed that participants are in compliance with applicable statutory and regulatory requirements for all open water activities, so as to prevent collisions at sea.

19 Specific open water sports

19.1 Additional information on the following open water sports is covered from Annexes G to R:

- a) Canoeing and kayaking;
- b) Dragon boating;
- c) Free diving;
- d) Improvised rafting;
- e) Open water swimming;
- f) Powered pleasure craft;
- g) Rowing;
- h) Sailing;
- i) Scuba diving;
- j) Snorkelling;
- k) Stand-up paddleboarding; and
- l) Water skiing and wakeboarding.

19.2 All stakeholders involved in the sport in one way or other, whether as an organiser, coach, participant, etc., shall comply with the relevant annex(es).

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Annex A
(informative)

Venue and equipment checklist

Event:

Date:

Table A.1 – Venue

S/N	Area	Opening √	Closing √	Remarks
1	<i>e.g. Female toilet</i>	☐	☐	<i>e.g. Remarks on sanitary conditions of the toilet.</i>
2	<i>e.g. Male toilet</i>	☐	☐	
3	<i>e.g. Lobby</i>	☐	☐	
4		☐	☐	
5		☐	☐	
6		☐	☐	
7		☐	☐	
8		☐	☐	
9		☐	☐	
10		☐	☐	
11		☐	☐	
12		☐	☐	
13		☐	☐	
14		☐	☐	
15		☐	☐	
16		☐	☐	
17		☐	☐	
18		☐	☐	
19		☐	☐	
20		☐	☐	
21		☐	☐	
22		☐	☐	
23		☐	☐	
24		☐	☐	
25		☐	☐	

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Table A.2 – Equipment

S/N	Item	Opening √	Closing √	Remarks
1	<i>e.g. AEDs</i>	☐	☐	<i>e.g. Remarks on working condition/quantity of the equipment.</i>
2	<i>e.g. First-aid kit</i>	☐	☐	
3	<i>e.g. PFDs</i>	☐	☐	
4		☐	☐	
5		☐	☐	
6		☐	☐	
7		☐	☐	
8		☐	☐	
9		☐	☐	
10		☐	☐	
11		☐	☐	
12		☐	☐	
13		☐	☐	
14		☐	☐	
15		☐	☐	
16		☐	☐	
17		☐	☐	
18		☐	☐	
19		☐	☐	
20		☐	☐	
21		☐	☐	
22		☐	☐	
23		☐	☐	
24		☐	☐	
25		☐	☐	

Staff in charge
(Opening): _____

Date: _____

Signature: _____

Staff in charge
(Closing): _____

Date: _____

Signature: _____

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Annex B (informative)

Checklist for safe water activities

S/N	Description	√	Remarks
1	Participant		
1.1	Age 1. Person under 12 years old is accompanied by a guardian aged 21 years and above 2. Water activities are done with a buddy, or next of kin has been notified of such activities 3. Elderly participants have a buddy for water activities	☐	
1.2	Health 1. Medically fit for activity 2. Heat acclimatisation	☐	
1.3	Competency 1. Ability to swim (PFD to be provided to non-swimmers and weak swimmers) 2. Comfortable with water activities (no aquaphobia/thalassophobia) 3. Experienced in specific water activity (e.g. stand-up paddleboarding (SUP), rafting, canoe, and kayak)	☐	
1.4	Participant is not under the influence of medication or alcohol	☐	
1.5	Participant is feeling well 1. Participant is not ill 2. Participant has sufficient rest and sufficiently hydrated 3. Participant is not anxious of water activities	☐	
1.6	Participant is attending the safety briefing before the activity	☐	
1.7	Participant is wearing the appropriate PFD, which fits snugly	☐	
1.8	Participant has protective gear 1. Closed toe footwear 2. Eye-protection (dependent on activity) 3. Sun protective gear or sunblock (dependent on activity)	☐	
1.9	Participant has an audible warning device, such as a whistle (optional)	☐	
1.10	Participant is aware of the emergency contact and procedures	☐	

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S/N	Description	√	Remarks
2	Environment		
2.1	Location is suitable 1. The boundaries are clearly demarcated 2. No incompatible activities are in the same area 3. The water condition is suitable for the water activity (e.g. no algae bloom, no shallow water with sharp stones, etc.) 4. Access to equipment and emergency services is provided 5. No known dangerous marine life in the area	☐	
2.2	Weather is suitable 1. No inclement weather (e.g. heavy rain, fog) 2. Tide and current is suitable for water activity 3. Sufficient sunlight for water activity (avoid 7 pm to 7 am)	☐	
3	Equipment		
3.1	Equipment (beach wheelchairs, paddle board, raft, canoe, kayak) are in good condition with no visible physical damage.	☐	
3.2	Oars and paddles are suitable for the specific activity (suitable length, suitable type – 1 paddle or 2 paddles).	☐	
4	Water activities		
4.1	Dry land practice	☐	
4.2	Emergency drills – Capsizes drills	☐	
4.3	The number of guides/spotters is sufficient for the activity.	☐	
4.4	Emergency equipment such float or emergency boat is deployed respective to the type of activity.	☐	
4.5	First aider is present for the activity.	☐	
4.6	First-aid equipment is available. 1. First-aid box 2. AED 3. Stretcher	☐	
5	Document		
5.1	Respective licences/permits are approved	☐	
5.2	Activity programs/schedule – Different groups	☐	
5.3	Certifications 1. Equipment certificates 2. Competency certificates	☐	
5.4	Risk assessment 1. Activity risk assessment 2. Environment risk assessment	☐	
5.5	Records 1. Personal declaration on health, competency, allergy, etc. 2. Training/briefing records	☐	
5.6	Incident reports 1. Near-miss incidents 2. Feedback/suggestions/recommendations	☐	

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Annex C (normative)

Management of hazards

C.1 General

The risk assessment plan shall include salient background information such as the following:

- a) Contact details of owners/operators/organisers;
- b) Description of locations including maps and diagrams of event course;
- c) Access arrangements for club, emergency vehicles, wheelchair users;
- d) Depth of water;
- e) Location of weed beds;
- f) Nature of water such as lake, river, sea, etc.;
- g) Average water temperatures for period of use; and
- h) Water quality.

C.2 Physical hazards

Key factors of physical hazards include the following:

- a) Access and egress, condition and proximity of start and finish points;
- b) Conditions underfoot;
- c) Current or eddies;
- d) Depth (changes);
- e) Equipment hazards such as inadequate, insufficient, faulty, ill-fitting, expired or wrong type of equipment, etc.;
- f) Hidden, overhanging or underwater hazards;
- g) Entrapment or entanglement hazards (e.g. ropes, activity prop);
- h) Water temperature – this can have implications on recommended swimming distance and immersions times;
- i) Other water users;
- j) Seasonal hazards (e.g. weeds); and
- k) Water quality.

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C.3 Hazards relating to participants

Key hazards relating to participants include the following:

- a) Participants with mixed ability/competency;
- b) Inexperienced open water participants; and
- c) Participants with low body fat that can lead to lower cold-water tolerance.

C.4 Marine animals

The common marine animals found in Singapore include the following:

- a) Barracudas and moray eels, which are biting fishes;
- b) Marine catfishes, which have venomous spines on their dorsal and pectoral fins;
- c) Pufferfishes, which are equipped with sharp, plate-like teeth that bite;
- d) Rabbitfishes and scats, which have venomous dorsal fin spines;
- e) Scorpionfishes such as stone fishes, which have a row of venomous spines on their dorsal fins; and
- f) Stingrays, which have two long venomous spines near the base of their tails.

NOTE – Refer to the Handline Fishing website for diagrams of marine creatures.

C.5 Water Quality

C.5.1 For recreational beaches, 95 % of the time:

- a) enterococcus count should be ≤ 200 per 100 ml of water;
- b) consider the susceptibility of the location to faecal influence; and
- c) only beaches classified as 'Good' and above can be considered suitable for primary contact activities.

C.5.2 For freshwater bodies, 95 % of the time:

- a) enterococcus count should be ≤ 200 per 100 ml of water; and
- b) chlorophyll-a concentration should be ≤ 50 mg per litre of water.

C.6 Microbial water quality assessment

A combination of sanitary inspection and microbial water quality assessment provides a basis for a robust and graded classification. Refer to Figure E.1 for an example of the method of classification.

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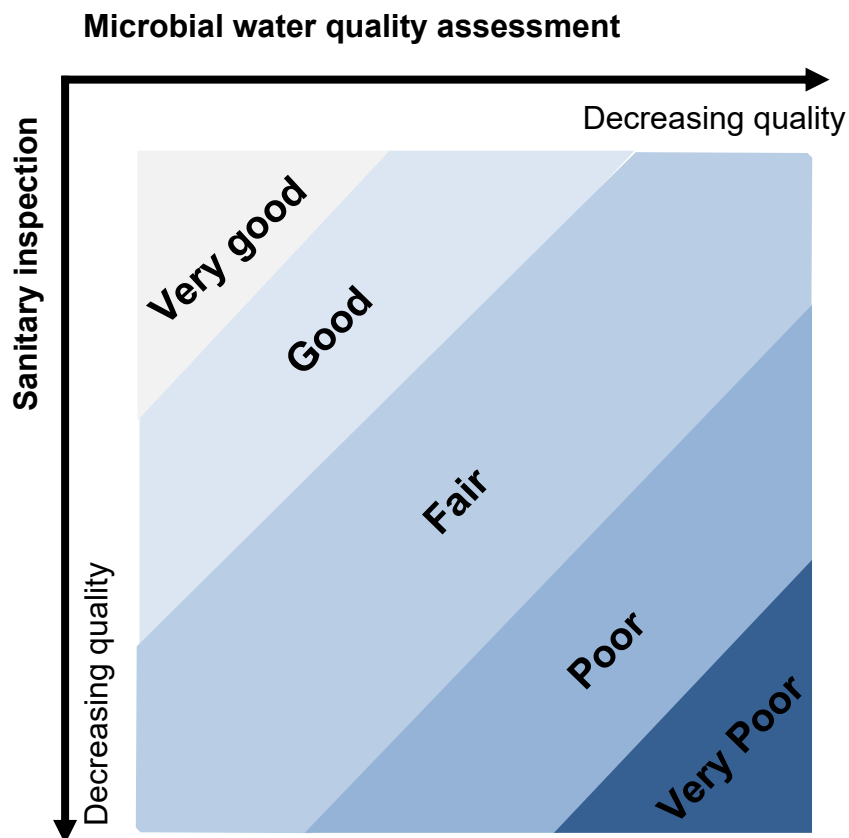


Figure C.1 – Method of classification

C.7 Algae bloom and risks

C.7.1 Algae blooms can occur naturally by the nutrient upwelling or other seasonal changes in the ocean. They can also be triggered by nutrient inflow into the sea from coastal pollution, such as farmland or untreated sewage, or by seasonal winds carrying particles from land to the sea. Algae blooms can be dangerous because the tiny algae produce toxins when ingested.

C.7.2 There is evidence that people who swim in or inhale such toxins can experience irritation in their eyes, nose and throat. Participants are likely to experience more severe symptoms if they have an existing respiratory illness such as asthma. Participants shall adopt the following measures:

- a) Avoid entering the water if foam, scum, or discolouration on the water's surface is sighted;
- b) Avoid potential exposure by rinsing oneself with clean, fresh water; and
- c) Seek medical treatment immediately if symptoms worsen.

C.7.3 If sailing, windsurfing, or undertaking any other activity likely to involve water immersion in the presence of algae blooms, participants should wear clothing that is close-fitting in the openings. All equipment shall be washed and dried after any contact with algae blooms and scum.

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C.8 Swimmer's itch or cercarial dermatitis

C.8.1 Swimmer's itch, also known as cercarial dermatitis, is a skin rash caused by an allergic reaction to certain microscopic parasites that infect some birds and mammals. These parasites can also burrow into a swimmer's skin, causing an allergic reaction. Within minutes to days after being in contact with contaminated water, participants can experience tingling, burning, or itching of the skin, along with small reddish pimples and/or small blisters. Scratching infected areas can lead to secondary bacterial infections, but the condition often subsides within a week without the need for medical attention.

C.8.2 Participants shall adopt the following preventive measures:

- a) Avoid swimming in areas where swimmer's itch is a known problem or where signs have been posted warning of unsafe waters;
- b) Refrain from swimming near or wading in marshy areas where snails are commonly found;
- c) Towel dry or shower immediately after leaving the water; and
- d) Avoid attracting birds (e.g. by feeding them) to areas where people are swimming.

C.8.3 Health officials shall make available information/signs on shorelines where swimmer's itch is common.

C.8.4 The following should be undertaken for relief of the condition:

- a) Use corticosteroid cream;
- b) Apply cool compresses to the affected areas;
- c) Bathe in Epsom salts or baking soda;
- d) Soak in colloidal oatmeal baths;
- e) Apply a baking soda paste to the rash; and
- f) Use an anti-itch lotion.

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Annex D (informative)

Risk assessment

Table D.1 – Examples of risk factors and its control measures

Subject	Risk	Controls
Activity for long duration (e.g. Marathon)	Heat exhaustion	Equip with the right attire Ensure frequent breaks and adequate hydration
Beach with concurrent activities	Colliding into swimmers	Demarcate and separate boating and non-boating activities
Participants	Non-swimmers	Avoid such water activity Have 1-to-1 guide
	Poor health	Avoid strenuous activity Have 1-to-1 guide
Training	Beginners	Conduct a thorough briefing Dry run on land Check on PPE
Emergency	Cuts	Provide first aid
	Drowning	Provide CPR

Table D.2 – Severity

Level	Severity	Description
5	Catastrophic	Death, fatal occupational disease or exposure, or multiple injuries.
4	Major	Serious injuries, serious occupational diseases, or exposure (includes amputations, major fractures, multiple injuries, occupational cancers, diagnosed mental illnesses, acute poisoning, disabilities and noise-induced hearing loss).
3	Moderate	Injury or ill-health (including mental well-being) requiring medical treatment (includes lacerations, burns, sprains, minor fractures, psychosocial stress, dermatitis and work-related musculoskeletal disorders).
2	Minor	Injury or ill-health (including mental well-being) requiring first-aid only (includes minor cuts and bruises, irritation, ill-health with temporary discomfort, fatigue).
1	Negligible	Negligible injury.

Table D.3 – Likelihood

Level	Severity	Description
1	Rare	Not expected to occur but still possible.
2	Remote	Not likely to occur under normal circumstances.
3	Occasional	Possible or known to occur.
4	Frequent	Common occurrence.
5	Almost Certain	Continual or repeating experience.

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Table D.4 – Risk evaluation

Risk level	Acceptability of risk	Recommended actions
Low	Acceptable	- No additional risk control measures are necessary. - Frequent review and monitoring of hazards are required to ensure that the risk level assigned is accurate and does not increase over time.
Medium	Moderately acceptable	- A careful evaluation of the hazards should be carried out to ensure that the risk level is reduced to As Low As Reasonably Practicable (ALARP) within a defined time period. - Interim risk control measures, such as administrative controls or PPE, may be implemented while long-term measures are being established. - Management attention is required.
High	Not acceptable	- High Risk level shall be reduced to at least Medium Risk before work starts. - There should not be any interim risk control measures. Risk control measures should not be overly dependent on PPE. - If practicable, the hazard should be eliminated before work starts. - Management review is required before work starts.

Table D.5 – Risk assessment matrix

Severity \ Likelihood		Likelihood				
		Rare (1)	Remote (2)	Occasional (3)	Frequent (4)	Almost Certain (5)
Severity	Catastrophic (5)	5	10	15	20	25
	Major (4)	4	8	12	16	20
	Moderate (3)	3	6	9	12	15
	Minor (2)	2	4	6	8	10
	Negligible (1)	1	2	3	4	5

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Annex E
(informative)

Boat checklist

Boat number: _____

Table E.1 – Safety boat inventory

S/N	Equipment	Opening √	Closing √	Remarks
1	Air horn	☐	☐	<i>e.g. Remarks on working condition/quantity of the equipment.</i>
2	Anchor	☐	☐	
3	Bailer	☐	☐	
4	Dry bag	☐	☐	
5	Fire extinguisher	☐	☐	
6	First aid kit	☐	☐	
7	Fuel tank and line	☐	☐	
8	Hand flare	☐	☐	
9	Insurance and certificate	☐	☐	
10	Kill switch	☐	☐	
11	Lifebuoy	☐	☐	
12	Paddle	☐	☐	
13	Patrol route map	☐	☐	
14	Propeller guard	☐	☐	
15		☐	☐	
16		☐	☐	
17		☐	☐	
18		☐	☐	
19		☐	☐	
20		☐	☐	
21		☐	☐	
22		☐	☐	
23		☐	☐	
24		☐	☐	
25		☐	☐	

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Table E.2 – Physical boat check

S/N	Item	Opening √	Closing √	Remarks
1	Engine	☐	☐	
2	Hull	☐	☐	
3	Inflatable tubes	☐	☐	
4	Propeller	☐	☐	
5	Transom board	☐	☐	
6		☐	☐	
7		☐	☐	
8		☐	☐	
9		☐	☐	
10		☐	☐	
11		☐	☐	
12		☐	☐	
13		☐	☐	
14		☐	☐	
15		☐	☐	
16		☐	☐	
17		☐	☐	
18		☐	☐	
19		☐	☐	
20		☐	☐	
21		☐	☐	
22		☐	☐	
23		☐	☐	
24		☐	☐	
25		☐	☐	

Safety Boat Officer
(Opening): _____

Date: _____

Signature: _____

Safety Boat Officer
(Closing): _____

Date: _____

Signature: _____

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Annex F (normative)

Shallow water blackout

F.1 General

Shallow water blackout, also known as hypoxic blackout, is a condition that can have severe and even fatal consequences.

The key to preventing hypoxic blackouts lies in understanding why individuals faint when exposed to breath-limiting, high-stress practices. The body's autonomic nervous system controls our breathing at an unconscious level, regulating our need to breathe based on the balance of oxygen and carbon dioxide (CO₂).

When CO₂ levels build up in the body, the response is to exhale the accumulated CO₂ and bring in fresh oxygen to nourish vital organs and muscles. However, when CO₂ levels are relatively low due to hyperventilation, whether intentional or due to strenuous activity, the brain loses its ability to signal the need for a breath. It mistakenly believes that there is not enough waste product built up to require breathing, even though the body needs that breath to survive.

Without adequate oxygenation, the brain begins to shut down, leading to a loss of consciousness. It can take as little as 2.5 min for brain damage and death to occur, compared to the recognised 6 to 8 min for a "normal" drowning submersion. This accelerated process is due to the body's already oxygen-deprived state and the warmth of the pool water.

F.2 Preventive measures

Coaches and swimmers shall take precautions to prevent shallow water blackouts when performing underwater drills. All training sessions should be overseen by a certified and experienced coach. It is important to acknowledge that nobody is immune to the risk of passing out in the water. Coaches and swimmers who use these drills should do so with caution and at their own discretion.

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Annex G (normative)

Canoeing and kayaking

G.1 General

G.1.1 A canoe is a lightweight, narrow water vessel, typically pointed at both ends and open on top. It is propelled by one or more seated or kneeling paddlers who face the direction of travel and use a single-bladed paddle.

G.1.2 Kayaking is the use of a kayak for moving over water, distinguished from canoeing by the sitting position and the number of blades on the paddle. A kayak is a low-to-the water, canoe-like boat in which the paddler sits facing forward, with their legs in front. Kayakers use a double-bladed paddle to propel the kayak, alternating strokes from side to side. Most kayaks have closed decks, although sit-on-top and inflatable kayaks are growing in popularity.

G.1.3 All participants should consider enrolling in certification courses to acquire the necessary skills and knowledge for canoeing and kayaking, as well as to adhere to safe practices.

G.1.4 In the event of unfavourable weather, the activity shall be cancelled.

G.2 Personal responsibility

G.2.1 Prior to the commencement of the activity, participants shall learn about the knowledge and risk assessment regarding matters such as the following:

- Identification of potential dangers around the activity location;
- Checking and examining carefully the tidal/current flow patterns;
- Water temperature; and
- Weather conditions.

G.2.2 Participants who are new to canoeing/kayaking should enrol in a canoeing/kayaking proficiency course and obtain certification as competent open water canoeists/kayakers.

G.2.3 Participants should be prepared for wet weather or the possibility of the canoe/kayak tipping over. It is advisable to bring extra clothing in waterproof bags/containers.

G.2.4 Participants should adopt a buddy system and never paddle alone, except for situations that prevent one from doing so or when an alternative equivalent solution is available. In case of inclement weather, everyone shall stay together to ride out the storm and support one another. Leave no one behind and paddle together as a group.

G.2.5 Participants shall take the following precautions:

- a) Always go paddling with safe, qualified, and experience canoe/kayak instructors who have the necessary qualifications and can conduct the activity safely in compliance with the authority's regulations.
- b) Always remain near to the coastline and avoid venturing too far into the open sea. If encountering strong currents, do not panic and stay on the surface until help arrives. Do not

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attempt to swim against the current. Be cautious of offshore winds, as they can make it challenging to return to shore.

- c) Be prepared by storing belongings such as food, drinks, clothes, map, first-aid kit, etc. in waterproof bags or containers. Participants can secure these waterproof bags/containers to one of the centre beams in canoes or kayaks to prevent loss if the canoe/kayak tips over.

G.2.6 Participants shall wear serviceable PFDs that fit their body sizes and body weight as specified by the manufacturer. It is important to check the weight-buoyancy ratio. Depending on the environment, appropriate attire should be worn. Sit-inside kayaks offer some protection from the elements, whilst sit-on-top kayaks result in more exposure to elements for users (e.g. to sun rays/UV light).

G.3 Organisers/operators/instructors/leaders/guides' responsibilities

G.3.1 Organisers, operators, instructors, leaders and guides shall have the necessary qualifications before conducting any programme.

G.3.2 Organisers, operators, instructors, leaders and guides shall conduct the following, among other, responsibilities:

- a) Prepare a risk assessment for the programme they are conducting;
- b) Conduct a safety briefing for all participants involved in the activity. This session shall cover essential information, such as:
 - objective of activity;
 - route;
 - proper usage of life jackets;
 - procedures for coping with capsized situations; and
 - dangerous zones/areas.
- c) Always practise and conduct capsize drills before the planned activity;
- d) Prepare an emergency and evacuation plan for open water activities;
- e) For open water expeditions, a safety boat equipped with communication devices shall be made available; and
- f) Consult and seek approval from the relevant authorities if there are plans to cross channels and fairways.

NOTE – In Singapore, the Maritime and Port Authority of Singapore (MPA) oversees the vehicular traffic in channels and fairways.

G.3.3 Organisers, operators, instructors, leaders and guides should plan for a reasonable instructor-participant ratio based on the following criteria:

- a) For a group of participants, some of whom have personal skills awards of 1-star and 2-star training in disciplines such as Canoe Polo, Canoe Sprint, Canoe Slalom, Canoe Marathon or any other discipline, the recommended ratio should be 1:10; and
- b) For a group of participants who have completed their personal skills award and training in inland waters (not open water), the ratio shall be 1:12. If there are additional safety guidelines in place (e.g. safety personal and boats on standby), the ratio should be 1:20.

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G.4 Getting into canoes/kayaks safely

G.4.1 To ensure a safe experience, participants shall wear PFDs before attempting to get into a canoe or kayak.

G.4.2 Participants should adhere to the following guidelines when getting into a canoe or kayak:

- a) Have someone hold it steadily to prevent unexpected falls or tip-overs.
- b) Crouch low with bent knees and hold the sides of the canoe or kayak for balance and move towards the seat.
- c) Walk along the centre of the canoe or kayak, keeping feet on the centreline to prevent rocking.
- d) Remain seated in the centre of the canoe or kayak to avoid tipping over; and
- e) Stay low and refrain from walking or standing in the canoe or kayak when away from shore. Sudden and jerky movements should be avoided as it can cause the canoe or kayak to tip over.

G.5 Canoeing/kayaking in open waters

To ensure a safe and enjoyable experience, participants should observe the following guidelines when canoeing or kayaking in open waters:

- a) Remain vigilant about water currents. If they notice the canoe or kayak being rapidly pulled by currents or if they spot rocks ahead in the waterway, participants shall immediately paddle away from the rocks or towards the shore;
- b) Stay away from low hanging trees and branches when near the shore.
- c) Avoid letting big waves hit the side of the canoe or kayak. They should always try to keep the canoe or kayak at a right angle to the waves to avoid potential tipping; and
- d) Refrain from canoeing or kayaking in adverse weather conditions that can be unsafe or undesirable for outdoor activities.

G.6 When canoes/kayaks tip over

Participants should observe the following guidelines when their canoes or kayaks tip over:

- a) Stay with the canoes or kayaks. They should paddle or push the canoes or kayaks to shore with the assistance of their buddy (i.e. the other person in their canoe); and
- b) In shallow water conditions, flip the canoes or kayaks to dump out the water and then re-enter the canoes or kayaks. These watercrafts are designed to remain buoyant even when filled with water.

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G.7 Low-powered engine kayaks

When using vessels with low-powered engines attached, kayakers shall:

- a) register with the relevant authorities before use; and
- b) be competent and licensed to drive power crafts in open seas.

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Annex H (normative)

Dragon boating

H.1 General

H.1.1 Dragon boats are human-powered watercraft categorised into two classes: 10-seater and 20-seater, referring to the number of participants in the boat. Participants sit in pairs side-by-side and use single-bladed paddles. In both classes, there can be drummers and steerspersons, with all boat occupants paddling to the rhythm of the drums.

NOTE – Refer to the Singapore Dragon Boat Association website for safety guides for dragon boating.

H.1.2 When fully compliant with safety requirements and conditions, dragon boats are considered safe for both non-swimmers and weak swimmers. The types of dragon boats and sizes of crew are listed as follows:

- a) 12-person dragon boats:
 - Maximum 10 participants + 1 trainer + 1 steersperson (total 12 individuals);
 - Maximum 10 participants + 1 trainer/steersperson (total 11 individuals for the non-novice team only);
 - Minimum 4 participants + 1 trainer + 1 steersperson (total 6 individuals); or
 - Minimum 4 participants + 1 trainer/steersperson (total 5 individuals for non-novice team only).
- b) 22-person dragon boats:
 - Maximum 20 participants + 1 trainer + 1 steersperson (total 22 individuals);
 - Maximum 20 participants + 1 trainer/steersperson (total 21 individuals for the non-novice team only);
 - Minimum 8 participants + 1 trainer + 1 steersperson (total 10 individuals); or
 - Minimum 8 participants + 1 trainer/steersperson (total 9 individuals for the non-novice team only); or
 - Minimum 8 participants + 1 trainer + 1 steersperson (total 10 individuals).

H.1.3 Dragon boat owners should refer to the dragon boat vessel inspection form when preparing their boats for inspections or repairs.

NOTE – Refer to the Singapore Dragon Boat Association website for the dragon boat vessel inspection form issued by PUB, Singapore's National Water Agency.

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H.2 Personal responsibility

H.2.1 All participants shall wear suitable PFDs that conform to ISO 12402-5, Conformité Européene (CE), European Standards (EN), or United States Coast Guard (USCG) standards. The wearing of PFDs shall adhere to the manufacturer's specifications and instructions. PFDs are rated by weight, so participants shall wear PFDs that are suitable to their body weight.

H.2.2 Participants should consider the following when deciding on the appropriate attire to wear:

- a) Clothing that can get wet without hindering free movement of the arms and legs;
- b) Clothing with sleeves to prevent possible abrasions on exposed skin caused by PFD straps and buckles;
- c) Clothes should be made of lightweight fabric to minimise additional weight, especially when in a capsized situation;
- d) Suitable footwear (e.g. shoes, sandals, or slippers) as the docking areas and the inside of the boat (the "deck") can become wet and slippery.
- e) Headgear such as caps and hats, if worn, should be fitted snugly and/or with chin straps to prevent them from being blown off; and
- f) Bespectacled persons should use spectacle straps or ear-hooks as a precaution in case their spectacles become dislodged due to the paddling motion.

H.2.3 Non-swimmers and weak swimmers shall inform trainers and steerspersons of their status before embarking.

H.3 Trainers' and steerspersons' responsibilities

H.3.1 Trainers shall be certified with the following qualifications, where appropriate:

- a) Singapore Dragon Boat Association (SDBA) Recreational Trainer/Instructor certificate; or
- b) SDBA Dragon Boat Technical Level 1 Coach certificate.

H.3.2 Steerspersons shall be certified with the following qualifications, where appropriate:

- a) SDBA Probationary Steering Licence; and
- b) SDBA Basic Helm certificate.

H.3.3 Trainers and steerspersons shall wear suitable PFDs that conform to standards such as ISO 12402-5, CE, EN or USCG standards. The wearing of PFDs shall adhere to the manufacturer's specifications and instructions. PFDs are rated by weight, so trainers and steerspersons shall wear PFDs that are suitable for their body weight.

H.3.4 Trainers and steerspersons shall conduct a safety and capsize drill briefing to all participants before boarding the dragon boat. Checks shall be made on all participants for:

- a) swimming skills/ability;
- b) appropriate attire and footwear; and
- c) appropriate and secureness of PFDs that are worn.

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H.4 Organisers' responsibilities

H.4.1 Organisers shall prepare risk assessment plans for the programmes they are conducting. Non-swimmers and weak swimmers should participate in the dragon boat orientation programme (DOP), which is conducted in inland waters (e.g. reservoirs).

H.4.2 When conducting programmes for schools, organisers shall include participants who are non-swimmers or weak swimmers in the outdoor adventure learning (OAL) programme, which is equivalent to the DOP.

H.4.3 When conducting programmes for schools under the sports education programme or in preparation for races, organisers shall ensure that all participants receive a safety briefing on the capsize drills and recovery drills.

H.4.4 Organisers shall deploy power/safety boats as a mandatory requirement for races. This requirement is optional for non-race activities but highly recommended for the safety of non-swimmers and weak swimmers participating in the activity.

H.5 Weather and water conditions

H.5.1 All activities on water shall be delayed, postponed or cancelled (dependent on time available) in the following situations:

- a) When authorities issue guidelines related to CAT 1 weather conditions and lightning warning alerts; and/or
- b) When conditions are deemed dangerous, as determined by trainers or steerspersons (e.g. high winds and waves, poor visibility due to heavy rain not classified as CAT 1, etc.).

NOTE – CAT 1 weather refers to heavy rain and/or lightning strikes (which can also occur without any rainfall).

H.5.2 During the onset of sudden dangerous situations, such as when there is lightning overhead, all dragon boats shall return to the base. All crew shall disembark and seek adequate and appropriate shelter on land. Alternatively, the crew can seek temporary shelter at the SDBA's designated temporary safe harbours.

NOTE – Refer to the SDBA website for the action plan during onset of weather conditions and the locations of safe harbours.

H.6 Embarkation and disembarkation

H.6.1 All participants shall fully cooperate and comply with the instructions and guidance provided by certified trainers and steerspersons, as they are trained to manage the embarkation and disembarkation of dragon boats. Embarkation and disembarkation shall take place either from the pontoon or from the beach.

H.6.2 Embarkation and disembarkation shall be done by using one of the following methods:

- a) One by one;
- b) Side by side; or
- c) Seat by seat.

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H.7 Buddy system

H.7.1 The buddy system assigns each participant with a partner for mutual checking on safety and support in an emergency, such as during a capsize or swamping. Trainers or steerspersons shall allocate a buddy to every participant. Refer to Figure H.1 for an example of how to allocate the buddy system among participants in a dragon boat.

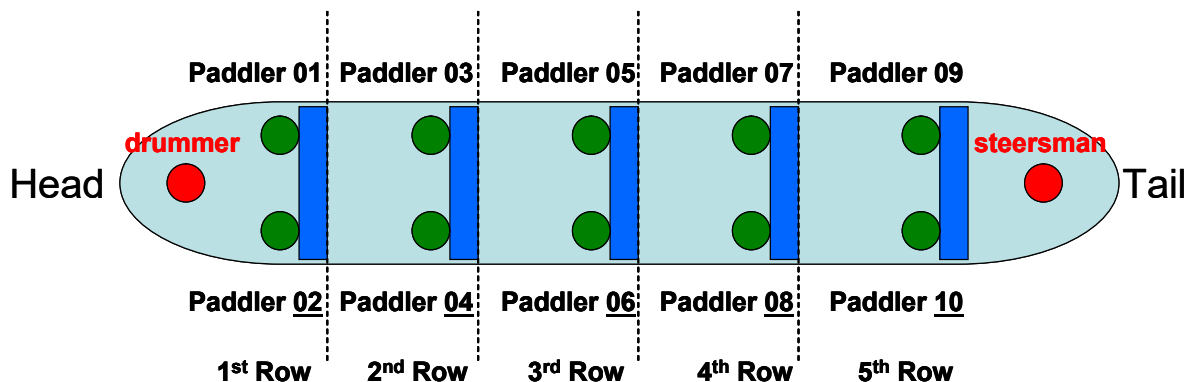


Figure H.1 – Example of buddy system

H.7.2 Participants shall be paired according to their seating positions. Drummers shall be grouped with the first row (first pair). The steersperson shall be grouped with the last row (last pair). The paired participants are now identified as ‘buddies’ for each other, with the first and last pair forming a three-person buddy system.

H.7.3 All buddies shall look out for each other’s well-being, signs of distress, or any visible negative physical conditions. In the event such situations arise, the buddy shall alert a trainer or steersperson.

H.7.4 In capsize or swamping situations, the participant shall first assess their own physical condition and, if there are no injuries, check on the physical condition of their buddy. If the buddy is physically missing or has missed a roll call, the participant shall alert a trainer, steersperson or first rescuer on site.

H.8 Capsize/swamping drills

H.8.1 Trainers and steerspersons shall ensure that participants are familiar with capsize and swamping drills. It shall be deemed unsafe to conduct a capsize drill with the actual dragon boat for novices as it involves technical knowledge and skills to recover a capsized boat. Therefore, more emphasis shall be placed on the theory segment. For enhanced safety, participants can request for simulated capsize drills or water confidence drills in a swimming pool on a different date before the actual dragon boat activity.

H.8.2 After a capsize or swamping incident, the crew can be ejected from the boat into the water. In such a situation, the crew shall do the following:

- Initiate self-preservation methods, such as keeping themselves afloat and maintaining an upright position immediately;
- Stay close to the capsized or swamped dragon boat and attempt to hold on to the hull. Dragon boats are designed with reserved buoyancy and do not sink immediately. Staying close to the boat facilitates the accounting of crew members as well as rescue efforts;
- Conduct a self-check for injuries and inform the trainers, steerspersons and their buddy if they have suffered any injuries; and

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- d) Check on the status of their buddies and inform trainers and steerspersons if their buddies are missing or too injured to account for themselves.

H.8.3 Trainers shall initiate a roll call to account for all crew members. If a crew member is missing, trainers should assume that the missing crew member is likely to be unconscious. They should conduct a quick visual search within their peripheral vision to locate missing paddlers. If the missing crew member is not within peripheral vision, trainers should assume they are under the overturned hull of the dragon boat. Trainers shall then initiate a leg sweep procedure under the boat to feel for anybody trapped there. Refer to Figure H.2 for the position of crew members on both sides of the capsized boat and Figure H.3 for an example of how to perform the leg sweep procedure.

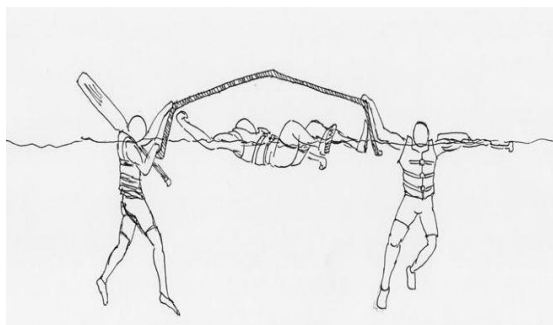


Figure H.2 – Position of crew members

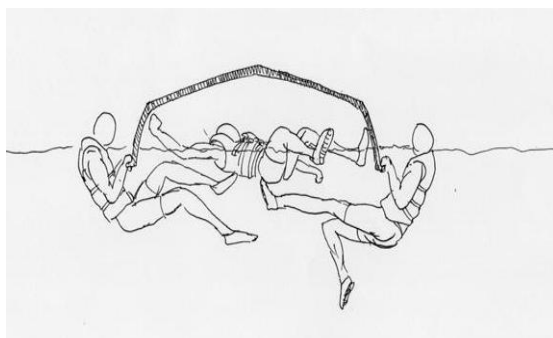


Figure H.3 – Leg sweep procedure

H.8.4 The ability to swim is not mandatory for dragon boating, but participants should have some exposure to swimming, although they do not necessarily need to pass swimming tests. In the event of an emergency, trainers shall immediately inform rescuers of any missing crew members and make an emergency call to emergency services.

H.8.5 When back on shore, the boat captain shall ensure that all crew members are accounted for. If any crew members are missing, the boat captain shall inform emergency services at the earliest possible time.

H.9 Recovery drills

H.9.1 Trainers shall assess whether it is possible to right the boat. If it is feasible, trainers shall attempt to right the boat by distributing the crew around the boat and holding on to it to prevent it from rolling over. No one shall swim under the boat to get to the other side, or swim around it.

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H.9.2 If trainers determine that it is not possible to right the boat, everyone shall do the following:

- a) Attempt to swim with the boat to the nearest shore, with the entire crew holding on to the boat and kicking to move with it; or
- b) Abandon the boat as per instructions from trainers and swim together to the nearest shore, with trainers and steerspersons keeping track of every crew member.

H.9.3 Upon reaching the shore, trainers shall conduct another headcount and call emergency services if anyone is missing.

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Annex I (normative)

Free diving

I.1 General

Free diving is a form of underwater diving that relies on breath-holding until resurfacing, without the use of breathing apparatus like scuba gear. In addition to the challenges of breath-holding, immersion in water and exposure to high ambient pressure also have physiological effects that limit the depths and duration possible in free diving.

I.2 Personal responsibility

Participants shall do the following:

- a) Develop plans for free diving activities, including risk assessments, identification of hazards/dangers at the activity venue, tidal and current flow patterns, weather conditions and emergency plans. During inclement weather, reschedule free diving activities for another day. Participants shall not mix free diving with scuba diving.
- b) Check sea conditions, dive sites, dangers and hazards, weather conditions, currents, visibilities, water temperatures, swell and surface traffic.
- c) Ensure they have enough rigging and flags.
- d) Dive with buddies and adhere to the rule of one up, one down.
- e) Avoid hyperventilation and consistently equalise their ears.
- f) Check surface interval times to ensure adequate recovery time for the next dive and abort all dives after a blackout incident, and
- g) Refrain from diving when fatigued and cold, and, most importantly, be mindful of their personal limitations.

I.3 Responsibilities of leaders/instructors/trainers

Leaders, instructors, and trainers shall adhere to the following:

- a) Possess knowledge, competence and experience in open water conditions as well as the training and leisure activities they provide. Their primary responsibility is ensuring the safety of training and leisure activities, and their secondary responsibility is ensuring suitable, safe open water environment are selected for these activities.
- b) Be aware of the following:
 - Maintain coach/trainer-trainee ratio of 1:4;
 - Understand the profiles and swimming capabilities/abilities of trainees and group them accordingly based on their capabilities;

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- Ensure that trainees complete pre-participation screening, such as the Get Active Questionnaire;

NOTE – Refer to the Sport Singapore website for an example of the Get Active Questionnaire.

- Conduct open water orientation programmes; educate participants about safety equipment usage and provide guidance on how to respond during emergencies; and
- Schedule regular breaks during activities and avoid venturing too far out into open waters.

c) Conduct safety briefings, detailing information such as the following:

- Objectives of dives;
- Buddy procedures, including in the event of lost buddies; and
- Safety procedures such as descent and ascent, confined waters/or open emergency drills, coping with strong currents, topside supervision, etc.

d) Supervise trainees closely and in full view of groups during descent, ascent and during training and leisure activities.

e) Conduct a debrief session after activities and maintain comprehensive logs of all information.

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Annex J (normative)

Improvised rafting

J.1 General

Improvised rafting is a human-powered activity that uses buoyancy flotation items tied or lashed together to improvise a watercraft to travel short distances across a water body. Rafts are flat structures used for support or transportation over water, typically kept afloat by using various buoyant materials and not propelled by engines.

J.2 Personal responsibility

Participants shall adhere to the following:

- a) Be aware of their own swimming ability and level of water confidence, and inform activity instructors if they are weak swimmers or have low water confidence;
- b) Ensure they are comfortable in the given water conditions while wearing properly fitted PFDs;
- c) Be capable of understanding and following safety instructions; and
- d) Equip themselves with the following attire and equipment, where appropriate:
 - Clothing and headwear that provide sun protection, along with appropriate footwear;
 - PFDs that are at least ISO 12402-5, SCF-endorsed or United States Coast Guard approved Type III certified; and
 - A whistle.

J.3 Responsibilities of organisers/conducting personnel/activity instructors

J.3.1 Organisers shall ensure that there is an adequate quantity of rafting equipment with the appropriate buoyancy for the intended weight and number of participants, as different materials have varying buoyancy and rigidity that can affect the stability of the rafts. In general, a stable raft should allow participants to sit on top of the structure, with their torsos above the water, and the raft structure should remain balanced and level.

J.3.2 Organisers shall have a thorough understanding of the properties and limitations of their equipment and ensure they are suitable for the water conditions in which the activities are conducted. This includes using the following:

- a) Wooden poles, bamboo poles, PVC pipes or kayaks for the frame;
- b) Inner tyre tubes and plastic barrels for buoyancy aid;
- c) Ropes or straps to lash the equipment together; and
- d) Sufficient paddles to propel the raft.

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J.3.3 Organisers shall plan activities in accordance with the following environmental factors:

- a) Launching and paddling of the raft shall not be done during night-time conditions;
- b) Construction of the raft shall be conducted on land;
- c) There shall be adequate visibility and lighting during the construction of the raft;
- d) Tidal and channel conditions shall be assessed for suitability of activities; and
- e) Weather forecasts shall be checked before the activity to anticipate inclement weather conditions, such as flash floods or thunderstorms (lightning risk).

J.3.4 Organisers shall adhere to the following practices when selecting activity locations for improvised rafting activities:

- a) Locations shall be approved by the respective authorities, owners or occupiers;
- b) Construction of rafts shall be located away from hazards (e.g. near launch points); and
- c) Appropriate launching and exit points shall be identified, considering factors such as swash, steepness, sharp rocks, slippery surfaces.

J.3.5 Organisers shall provide appropriate safety escorts and qualified activity instructors based on venue locations and conditions, as shown in Table J.1.

Table J.1 – Safety support for improvised rafting in different open water environments

Environment	Minimum level of safety escort	Ratio of safety crafts to raft	Rescue plans for common incidents (e.g. raft breaking apart; man overboard; raft drifting to danger; sudden bad weather)	Licences and safety competencies
1. Open water (Coastal) – wind, tide, waves, temperature	For distances beyond 20 m from shore – motorised craft	1 rescue craft to max. 1 raft, or 1 motorised boat to max. 2 rafts	- Tow raft to safe areas - Pick up & transfer pax out of water quickly	- PPCDL - CPR-AED (SRFAC) - Standard FA - Water recovery and rescue skills
	For distances less than 20 m from shore – no safety craft required, provided Activity Instructor is onboard the improvised rafting.	1 Activity Instructor onboard 1 raft, or 1 rescue craft to max. 1 raft, or 1 motorised boat to max. 2 rafts	- Maintain < 20 m distance from shore - Take direct charge to steer the raft - Assist participants to swim/row toward nearest put-out point	- Kayak Proficiency Level 2 Star - CPR-AED (SRFAC) - Standard FA - Water recovery and rescue skills

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Table J.1 – Safety support for improvised rafting in different open water environments (continued)

Environment	Minimum level of safety escort	Ratio of safety crafts to raft	Rescue plans for common incidents (e.g. raft breaking-apart; man overboard; raft drifting to danger; sudden bad weather)	Licences and safety competencies
2. Waterways and lagoons (no more than Grade 1) – some wind, some tide, obstructions in water	For distances beyond 20 m from designated beach up point – motorised craft For distances less than 20 m from designated beach up point – no safety craft required, provided Activity Instructor is onboard the improvised rafting.	1 rescue craft to max. 1 raft, or 1 motorised boat to max. 2 rafts 1 Activity Instructor onboard 1 raft, or 1 rescue craft to max. 1 raft, or 1 motorised boat to max. 2 rafts	• Push raft out of hazardous areas • Direct raft to middle of waterway channel • Assist participants to swim/row toward nearest put-out point	• CPR-AED (SRFAC) • Kayak Proficiency Level 2 Star • Standard FA • Water recovery and rescue skills
3. Inland lake (quarry, reservoir) – freshwater, placid, deep water; entanglements in water	Instructor on board raft	1 rescuer to max. 1 raft	• Maintain < 20 m from shore • Take direct charge to steer the raft • Swim to shore if capsized	• CPR-AED (SRFAC) • Standard FA • Water recovery and rescue skills
4. Swimming pool	Instructor on the shore	1 rescuer to max. 20 participants	• Assist participants to swim/row toward nearest pool edge • Deploy rescue means (pole, buoy, throwline)	• CPR-AED (SRFAC) • Standard FA • Water recovery and rescue skills
NOTE – Rescue craft refers to kayaks, canoes, inflatable boats and other non-motorised but stable paddle craft suitable for towing a raft during a rescue in the water.				

J.3.6 In Singapore, conducting personnel and activity instructors shall minimally possess the following relevant certifications for the activities:

- Singapore Resuscitation and First Aid Council Standard First Aid;
- Singapore Resuscitation and First Aid Council CPR and AED certification; and
- Singapore Canoe Federation Kayak Proficiency Level 2 Star or equivalent (where kayaks are utilised as safety craft).

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J.3.7 Conducting personnel and activity instructors shall be competent in the following raft construction and water recovery and rescue skills:

- a) Supervising up to 20 participants in the water;
- b) Assessing the water-readiness condition of a raft;
- c) Working in the operating area for the activity, including identifying hazards, evacuation points and routes;
- d) Demonstrating proficiency in ropework, including knots and lashings relevant to construction (e.g. clove-hitch, square lashing);
- e) Executing the lifting and flipping techniques of a raft;
- f) Instructing participants on how to board and disembark from an improvised raft;
- g) Proper steering and paddle strokes including ruddering, forwards and backwards moving, stopping, and turning;
- h) Assessing environmental hazards (e.g. strainers, lightning and tidal conditions);
- i) Raft recovery and/or towing techniques if the raft breaks up or drifts toward danger;
- j) Managing the break-up of the improvised raft while in the water; and
- k) Rescuing participants from entrapment caused by equipment used in an improvised raft.

J.3.8 Activity instructors shall do the following:

- a) Conduct risk assessments for activities; and
- b) Equip themselves with appropriate attire and equipment, such as:
 - Clothing and headwear that provides sun protection, including appropriate footwear;
 - Appropriate safety craft for the activity site, environment, and distance from shore – which can include kayak, canoe, inflatable rowboat, or a motorised vessel. For activities within 20 m from the shore, whether in open waters or flat-water bodies, safety crafts are optional, provided the conducting personnel or activity instructor is onboard the raft. For activities in swimming pools, safety crafts are optional and the conducting personnel or activity instructor can be onshore (refer to Table J.1);
 - Communication devices (e.g. walkie-talkies, mobile phones, and VHF radios);
 - Signalling devices (e.g. whistles and air-horns);
 - Cutting tools (e.g. knives and shears for rope entanglement); and
 - PFDs that conform to standards such as ISO 12402-certified Type III flotation devices.

J.3.9 Other rescue equipment shall also be included depending on rescue plans (e.g. throw lines, tow lines and spare paddles).

J.3.10 Pre-activity checks by activity instructors shall ensure the following:

- a) The location is free of hazards;
- b) Participants are physically ready to participate; and

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- c) Materials and paddles are safe for use.
- Materials used to construct the frame of the improvised raft have no sharp or split edges or surfaces;
 - Buoyancy aids to be attached to the improvised raft are not cracked, damaged or leaking;
 - Ropes/straps or material to hold the parts of the improvised raft together are sufficient, not frayed and are adequately strong to hold the raft together;
 - Paddle blades are firmly attached to the shaft; and
 - Paddles have no sharp or split surfaces.

J.3.11 Pre-launch briefings by activity instructors shall include the following:

- a) Conducting a physical check (“lift-and-shake test”) to assess the water-readiness of the improvised raft;
- b) Ensuring that loose ends of ropes are tied down and other potential entanglement hazards are removed;
- c) Providing participants with information on the risks and relevant rescue plan, including what to do if a person falls overboard, drifting, bad weather and the break-up of the raft;
- d) Briefing participants on how to mount and dismount an improvised raft;
- e) Warning against dangerous behaviour, such as performing dives or acrobatic acts while on the raft;
- f) Teaching participants directional control of the improvised raft and proper use of paddles for forward and reverse movement, as well as turning the raft on the spot while it is afloat; and
- g) Briefing participants on safety procedures to follow when the improvised raft breaks apart, emphasising the importance of staying close together and hanging on to the pipes and drums
 - no one should swim away from the group, unless they are approaching a hazard.

J.3.12 Activity instructors shall monitor the construction process of rafts and guide participants in the design and proper application of knots and lashings, if necessary.

J.3.13 Activity instructors shall assess the rafting areas for any environmental hazards, consider weather forecasts and tidal conditions, and ensure that activities are not conducted during heavy rain, thunderstorms or under conditions of reduced visibility.

J.4 Launch of raft/beaching-up phases

Activity instructors shall do the following when launching the raft or beaching up:

- a) Conduct water confidence tests with participants using PFDs in shallow waters prior to the first launch.
- b) Be aware of participants’ swimming abilities and buddy weak swimmers with stronger ones, where possible.
- c) Ensure participants wear their properly fitting PFDs that are securely fastened at all times.

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- d) Determine appropriate and safe launch sites and landing sites. When launching from or landing at pontoons, trainers/instructors shall:
 - keep a lookout to ensure participants do not fall into the water and are swept under pontoons; and
 - be aware of the gradient of the beaches/banks and hazards associated with shore-breaks.
- e) Brief participants on how to launch/beach up rafts. During the process, participants shall avoid standing within rafts as well as between rafts and beaches to avoid getting hit by rafts carried by waves.

J.5 During activities

J.5.1 Activity instructors shall ensure all participants and their rafts remain within sight and have quick access.

J.5.2 Activity instructors shall constantly monitor the speed and direction of tidal currents and winds, floating debris, dangerous marine life and other moving boats/ships, and direct participants if necessary.

J.5.3 If the improvised raft breaks apart and capsizes, activity instructors shall prioritise the safety and well-being of all participants before retrieving any equipment. This includes immediately accounting for all participants and recovering them from the water within a reasonable time.

J.5.4 Activity instructors shall conduct an immediate headcount and ensure that no one is entangled by the ropes.

J.5.5 Activity instructors should instruct participants to swim to shore (if close enough); or await rescue by safety boats/safety kayakers. No one shall swim away from the group.

J.5.6 In the event rafts are recalled, participants shall immediately paddle towards the shore.

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Annex K (normative)

Open water swimming

K.1 General

Open water swimming can take place anywhere that is not a swimming pool. This means no man-made sides and bottom, and no lane ropes to follow. Open water bodies include (but are not limited to) lakes, rivers, seas and reservoirs.

K.2 Personal responsibility

K.2.1 Participants shall do the following:

- a) Know their own abilities and capabilities before venturing into open water. If in doubt regarding their ability to handle water conditions, participants shall not enter the water;
- b) Check entry and exit points, and understand currents and tides;
- c) Do not swim if they have consumed or are currently consuming alcohol;
- d) Buddy up and swim with people who know the area and are familiar with swimming in open waters. Inform someone of your swimming location;
- e) Enter the water slowly to prevent “cold water shock”. They shall not jump off rocks and water breakers, nor run and dive into waters. Participants shall not swim at night; and
- f) Be visible in waters. They shall wear appropriate attire (e.g. wearing brightly coloured cap, and consider using tow-floats).

K.2.2 If someone is in trouble, participants shall not put themselves at risk, but call for help. If participants are in trouble, float to survive.

K.3 Potential dangers

Participants shall be aware of potential dangers such as the following:

- a) Moving watercrafts (e.g. canoes, boats and jet skis);
- b) Waves, including plunging, surging, and spilling waves;
- c) Currents, including tidal currents, runback currents and rip currents (fast and strong). Rip currents can sweep participants into deep waters (away from shore);

NOTE – Rip currents can be identified with debris floating with the currents; discoloured water due to the sand being stirred; and waves breaking on either side of the rips. If caught in rip currents, the participant is advised not to panic, swim parallel to the shore until free of the currents or swim back to shore after the currents break apart.

- d) Water currents moving towards the sea; and
- e) Rocks.

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Annex L (normative)

Powered pleasure craft

L.1 General

L.1.1 The term “powered pleasure craft” refers to a pleasure craft propelled by machinery. No person shall drive a powered pleasure craft unless they hold the appropriate class of pleasure craft driving licence as specified in Table L.1.

Table L.1 – Classes of pleasure craft driving licences

Length	< 24 m	≥ 24 m
Class of pleasure craft driving licence required	Powered pleasure craft driving licence	Advanced powered pleasure craft driving licence

L.1.2 No person shall use a pleasure craft, or cause or permit a pleasure craft to be used, within the port without a licence. A pleasure craft may be licensed either for private use or for commercial use. The different categories of registration in Singapore are as follows:

- a) SZ licence: These are pleasure craft for private use. “Private use”, in relation to a pleasure craft, means the use of the pleasure craft within the port by its owner or his guests for sport or pleasure purposes, without any such guest having to pay any fee or any other form of consideration.
- b) SZH licence: These are pleasure craft for commercial use and may only operate within the Singapore Port Limits. “Commercial use”, in relation to a pleasure craft, means the offer of the pleasure craft by the owner thereof to any person for hire or charter for that person’s use for sport or pleasure purposes within the port.

L.1.3 All pleasure craft plying beyond the Port of Singapore and in the high seas shall be registered with a ship registry. Pleasure craft registered with the Singapore Registry of Ships (SRS) shall be exclusively used by the owner, or a guest of the owner, for sport or pleasure purposes only. Registered pleasure craft shall not be used for any commercial purposes.

L.1.4 Pleasure craft registered with a ship registry but do not have a pleasure craft permit (i.e. “SZ” or “SZH”), are required to apply for a cruising permit before they are allowed to cruise within the port.

NOTE – Refer to the MPA website for the cruising permit and for more information on this permit.

L.1.5 Harbour craft and regional ferry services are not covered as this standard does not include these commercial activities.

L.2 Driving a powered pleasure craft

Skippers of powered pleasure craft cruising in the Port of Singapore shall hold a valid Powered Pleasure Craft Driving Licence (PPCDL for vessels up to 24 m), or an Advanced Powered Pleasure Craft Driving Licence (APPCDL for vessels above 24 m) issued by the Port Master.

NOTE – Refer to the MPA website for more information on licensing of pleasure craft.

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L.3 Personal responsibility

L.3.1 Persons operating a powered pleasure craft should demonstrate proficiency in identifying local islands, buoys, beacons, and other navigational aids that enhance their situational awareness while on the water

L.3.2 Persons operating a powered pleasure craft shall:

- a) practise and uphold the principles of good seamanship. This entails fostering a profound appreciation and respect for the sea, as well as demonstrating a sincere regard for the well-being and safety of fellow seafarers. Furthermore, they should engage in practicing boating etiquette, thereby ensuring a harmonious and respectful environment for everyone on the water.
- b) know their location and intended course when at sea. They may use an appropriate scale paper chart, or an electronic chart system designed to assist with navigation.
- c) maintain a vigilant lookout at all times. They shall take prompt and decisive actions to avoid collision and not impede the safe passage of vessels along fairways. Fairways are the main throughfare of shipping in a harbour and are usually marked by lateral buoys leading to and from a wharves or jetties.
- d) not assert or insist on the right of way, particularly when interacting with ships or commercial vessels, which are generally less manoeuvrable than powered pleasure crafts.

L.3.3 It is crucial to recognise the inherent dangers of drowning at sea. The marine environment significantly differs from that of a controlled swimming pool, characterised by the presence of waves, strong winds, tidal streams, overfalls, eddies, and ocean currents. Even proficient swimmers can experience rapid fatigue in such conditions. Therefore, in adverse weather conditions or when uncertain about the prevailing circumstances, all persons on board the craft should wear a lifejacket or PFD. This precautionary measure helps ensure their safety and survival. Children aged 12 and below should wear a lifejacket when on the open deck of a powered pleasure craft, as an additional safety measure to protect their well-being.

L.3.4 The skipper of a powered pleasure craft shall prioritise the safety of the boat, crew, and guests and the environment.

L.3.5 The pleasure craft, regardless of its size, is one's best life raft. If the craft remains afloat, staying with it should be the default choice, even if it is disabled, taking in water or has experienced a small fire. Remaining with one's vessel is generally preferable to relying on a life raft, life buoy or lifejacket anytime. It makes it easier to be spotted by rescuers while maintaining access to food, water, warm clothing, navigation equipment, VHF radio, a boat horn and flares.

L.4 Equipment

It is presupposed that the equipment carried by pleasure craft are in compliance with applicable statutory and regulatory requirements.

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Annex M (normative)

Rowing

M.1 General

Rowing is the act of propelling boats forward in water using the motion of using fixed oars as a lever to displace water.

M.2 Personal responsibility

M.2.1 Participants shall do the following:

- a) Strictly follow the defined traffic pattern. As rowing involves facing backwards while moving forward, all participants shall follow a similar movement pattern to prevent collisions;
- b) Be mindful of water conditions, such as choppy waters, which can increase the risk of boats capsizing as compared to calm water surfaces; and
- c) Learn proper techniques for handling the boat on and off the water, as well as any other equipment related to the sport.

M.2.2 Rowers using riggers to support the oars should be aware that the sliding motion of rowing can pose a risk if their PFD becomes stuck. In addition, in the event of a capsize, the PFD can hinder the rower's self-rescue efforts. Therefore, all rowers shall have basic swimming skills and undergo capsize drills. The PFD shall be kept in the rowboat.

M.3 Requirements for boats and crafts

M.3.1 Boats shall be fitted with white balls of at least 4 cm in diameter. They shall be made using supple or plastic. They come as standard in rowing boat construction and help prevent major injuries in the event of a collision on or off the water. The hull of boats shall not have any holes or cracks as this can affect their buoyancy. The shaft of oars shall not have any holes or cracks that can affect their buoyancy.

M.3.2 For motor-powered crafts, their engines shall be in good working condition. Their hulls shall not have any holes or cracks that can affect their buoyancy; and their outboard engines shall be secured to their hulls.

M.3.3 Umpiring crafts shall be motor-powered. These crafts shall be equipped with warning tools and the hulls shall not have any holes or cracks that can affect their buoyancy.

NOTE – For rowing in reduced daylight, the World Rowing Federation's "Minimum guidelines for the safe practice of rowing" states that the crafts are fitted with lightings.

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Annex N (normative)

Sailing

N.1 General

Sailing, as with any other water sports, can be safe and enjoyable but dangerous if safety is ignored. From novices to experienced sailors, it is vital to exercise safety on waters.

NOTE – Refer to the Sport Singapore website for the safe sailing guide.

N.2 Personal responsibility

N.2.1 Participants shall adhere to the following principles:

- a) Have a clear understanding of their own limitations and capabilities. If feeling unwell and/or on medication, participants shall refrain from sailing. It is essential to always respect the sea and sail within their limits. When in doubt, refrain from sailing.
- b) Check weather forecasts, tides, and currents before each sailing session. Participants shall be prepared to adjust their schedules or alter their course midway through the sail, as tidal streams can be very strong during “third hour tide”.
- c) Stay alert to any changes to weather conditions, such as approaching storm clouds. When anticipating inclement weather, it is highly advisable to sail closer to shore and be ready to seek shelter when a storm hits. Common storm risks include lightning, strong gusts of wind and poor visibility.
- d) Be familiar with areas where activities are conducted. Sail at recognised clubs or locations and inform someone of their planned activities.
- e) Avoid obstacles and dangerous areas, including rocks, strong currents, and sea lanes. Participants shall obey sailing and maritime rules to prevent collisions.
- f) Keep an eye out for breaking waves when holding boats at the shore. These waves can push the boat against participants and cause injury.
- g) Always practise capsizing procedures in a controlled environment.
- h) Be aware of marine litter (e.g. rusty nails, broken glasses and fishing hooks) as these can cause injuries, especially when the water is murky and visibility is poor.
- i) Seek immediate medical attention at the nearest clinic or hospital if they come into contact with or are stung by sea creatures, whether near the shore or at sea.
- j) Apply waterproof dressing to existing cuts and wounds before sailing, as open waters can contain bacteria.
- k) Use/wear personal protective gear/attire, such as the following:
 - Sunglasses, hats or cap;

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- Gloves, which are highly recommended to prevent rope burns (friction burns to hands) while managing ropes during sailing;
- Approved PFDs with a mirror and whistle, which shall remain functional even when wet. The PFDs should be both useable and properly fitted, with the whistles loud enough to be heard over choppy waters;
- Long-sleeved T-shirts and trousers; and
- Proper footwear, such as booties (not slippers) when in the water or onshore.

N.2.2 Participants should use sunblock and/or lip-protection balms to protect themselves during prolonged outdoor exposure.

N.2.3 Before sailing, participants shall perform the following:

- a) Inspect boats for damaged or missing parts. Replace any missing parts and report any to the person-in-charge. This process shall also be repeated at end of every sail.
- b) Check buoyancy compartments for any damage, as they keep boats afloat in the event of a capsizing.
- c) Ensure all ropes and riggings are properly secured, as loose masts may cause injuries.
- d) Secure all personal items (e.g. water bottles, snacks) in waterproof bags, and
- e) Follow all safety instructions provided by the person-in-charge.

N.3 When launching

Participants shall do the following:

- a) Adhere to all safety procedures as instructed by the person-in-charge; and
- b) Take extra care when boarding boats as they are prone to capsizing. Step as far as possible into the centre of the boat and never step on their edges.

N.4 During sailing

Recreational participants should observe the “right of way” between sailing boats and familiarise themselves with the racing rules of sailing (RRS) which include the following:

- a) Avoid collisions with other boats, crafts, buoys, beacons, and other objects. Participants should remain vigilant and keep a lookout for other boats and navigational structures. Participants should give other boats ample space before tacking or gibing when in close proximity to other structures.
- b) Hold on to something and be prepared for potential dangers. Large waves or sudden gusts of wind can toss boats without warning. Participants should always bend their knees for better balance, and
- c) Immediately deploy life belts and into the waters. If participants themselves end up in the water, try to maintain visual contact with the boat. Use whistles and mirrors to get the crew’s attention.

NOTE – Refer to the World Sailing website for the RRS. The RRS governs the sport of sailboat and sailboard racing, and is revised and published every four years.

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Annex O (normative)

Scuba diving

O.1 General

Scuba diving is a popular underwater recreational activity for participants (both beginners and experienced ones) to explore the underwater coral environments in the open sea. Diving courses are available for beginners, conducted by qualified instructors. Dive leaders, instructors, trainers, and participants shall exercise due care to ensure safe diving practices.

O.2 Personal responsibility

O.2.1 Participants shall ensure the following:

- a) Complete certified diving courses and be certified as open water divers before undertaking any open water dives.
- b) Always dive with safe, qualified, and experienced dive leaders who can conduct activities in compliance with the relevant authority's regulations, diving safety procedures, standards and protocols.
- c) Never dive alone; adopt the buddy system. Participants shall look out for their diving buddies and observe for "silent vulnerable" participants who are likely to freeze underwater, and practise and establish effective hand signals and communication. Participants shall be proficient in underwater emergency drills;
- d) Wear a serviceable, high-quality buoyancy compensator that fits their body size and weight, as specified by the manufacturer.
- e) Be current and depth-sensitive; learn how to manage surface and underwater currents. Ascend and descend safely, keeping to planned diving parameters. In the event of strong currents, participants shall practise the "stop, think, breathe, regain composure" sequence and stay safe by moving horizontally with the current until reaching a point of slack current. If currents are pulling participants downward, they shall ascend by dropping their weight belt, kicking upward, inflating the buoyancy compensator, and escaping the swirling downward currents (caused by underwater earthquake/movements).
- f) Be aware of their body conditions. If they feel tired and breathless, participants shall practise the "stop, think, breathe, regain composure" sequence and signal to their buddy or dive leader/instructor/trainer. Tiredness and breathlessness can result from breathing contaminated air from scuba tanks, while shallow water blackout risks can result from hyperventilation or elevated carbon dioxide levels (refer to Annex F).
- g) Assess their personal skills, physical, medical, mental, and emotional fitness prior to any diving activity.
- h) Ensure that they are briefed on details covering dive site details, dive procedures, estimated dive time, emergency procedures, potential unusual hazards or environmental conditions that can affect dive safety.
- i) Check that the equipment they are using is in good working condition to the best of their knowledge and ability.

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- j) Follow the instructions provided by the dive leader/instructor/trainer, including dive depth, duration, actions to be taken during distress and avoidance of prohibited activities in water, if any.
- k) Report any sickness or discomfort to the dive leader/instructor/trainer before and/or during the activity.
- l) Complete the fitness to dive questionnaire provided by the organiser before commencing a dive. Participants with medical issues as highlighted in the questionnaire should obtain certification from a medical examiner of divers within 1 year of the dive. Participants aged 45 and above should be certified by a medical examiner of divers and have an electrocardiogram examination performed within 1 year before the dive.
- m) Ensure that all relevant details of the diving activity are recorded in their logbook.

O.2.2 In the event of inclement weather, participants shall stay together to ride out the storm and provide mutual support. They shall ensure that no one is left behind and stay together as a group. In high wave situations, use regulators or snorkels at the water's surface. Participants shall avoid swimming against strong currents and high waves. Instead, they should stay afloat until help arrives.

O.3 Participants' competency

O.3.1 A certified open water diver is an entry-level diver and is competent to use all appropriate scuba diving equipment for open water diving in a safe manner. They shall meet the following requirements:

- a) Be of minimum age: adult divers are those who are 15 years and above, while junior divers are aged between 10 to 14 years. Certified junior divers shall be accompanied by an adult diver buddy; and
- b) Completed 4 logged open water dives at a maximum depth of 12 m under the supervision of a certified dive instructor.

O.3.2 Certified participants shall have basic knowledge of, but not limited to, the following:

- a) Diving physics (an understanding of gas laws, underwater pressure, compression/decompression of gas, which can affect ear and sinus equalisation, breathing, underwater performance and diver buoyancy);
- b) Physiological effects of diving on the human body and diving-related health issues, such as squeeze and overexpansion injuries, nitrogen narcosis, decompression sickness, and CO or O₂ poisoning;
- c) Scuba diving systems and equipment, including masks, snorkels, fins, buoyancy compensators, weight belts, demand regulators, submersible pressure gauges (SPGs), depth gauges, dive compasses, dive computers, and alternate air sources such as the second stage regulator mouthpiece or a separate spare air gas bottle;
- d) Diving planning and the use of no-decomposition dive tables to prevent decompression sickness, as well as ensuring sufficient rest and dive time;
- e) Maintaining diving records and logging open water dives;
- f) Understanding the underwater environment, recognising hazards, dangers and risks, and knowing how to manage them;
- g) Different types of diving activities; and
- h) Dangerous marine animals and measures to stay clear of and avoid them.

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O.3.3 Certified participants shall have the following practical skills:

- a) Swim continuously for 200 m using any stroke;
- b) Demonstrate survival swimming skills (treading, floating and/or drown-proofing) for 10 min;
- c) Swim underwater for a minimum distance of 15 m in one breath without diving in or pushing off;
- d) Execute proper techniques for entries, exits, descents, ascents, surface dives, surface swimming, clearing the snorkel, weight belt removal and replacement, buoyancy control, air supply planning, underwater swimming, performing skin diver rescues and executing underwater emergency buddy air sharing rescues;
- e) Demonstrate scuba diving skills with proficiency, including underwater breathing, mask and ear clearing, and mouthpiece clearing and retrieval drills;
- f) Understand the selection, checking, assembly, adjustment and donning of equipment; able to perform pre-dive gear checks for themselves and their respective buddies, defog masks, properly doff equipment after diving, and rinse and care for gear as needed;
- g) Give and recognise surface and underwater communication using hand signals;
- h) Plan and execute a no-decompression dive to a depth of 12 to 18 m, ensuring adequate air for descent, time at depth, ascent, precautionary stops, and safety margins; and
- i) Recover a diver simulating unconsciousness to the surface from a depth of approximately 3 m, remove the victim's weight system, mask and snorkel, and simulate in-water rescue followed by a surface tow of the victim for at least 20 m.

O.4 Equipment

O.4.1 Participants shall ensure the following:

- a) Their personal scuba regulators (including the mouthpiece) are inspected and tested prior to first use and every 12 months thereafter, and to check for any sign of wear or deterioration of rubber-based items.
- b) The mouthpiece of regulators they are using for the dive are in good condition and firmly secured.

O.5 Responsibilities of dive leaders/instructors/trainers

O.5.1 Dive leaders, instructors and trainers shall adhere to the following:

- a) Be knowledgeable, competent, and experienced in open water conditions as well as the training or leisure activities being provided. This includes ensuring:
 - the activity is safe; and
 - the selection of a suitable, safe open water environment for the activity.
- b) Be cognisant of the following:
 - Maintain a small coach/trainer-trainee ratio 1:4;

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- Adhere to staffing ratios for adult participants and advanced participants, as indicated in Table O.1;
- Understand participants' profiles and swimming capabilities/abilities, and group participants with similar capabilities together;
- Ensure participants complete pre-participation screening, including the Get Active Questionnaire and the medical declaration form for scuba diving training;

NOTE – Refer to the Sport Singapore website for an example of a Get Active Questionnaire.

- Conduct an open water orientation programme that cover the use of safety equipment and emergency procedures; and
- Incorporate regular breaks during activities and avoid venturing too far out into open waters.

Table O.1 – Staffing ratio for certified adult participants

Participant's experience	Dive team shall be accompanied by	Maximum number of participants in the dive team (including the dive leader)
≤ 10 dives	An experienced leader or dive master	6
> 10 dives	A dive leader	4

NOTE – A dive leader is chosen by the dive instructor or master, or the dive team.

- c) Develop dive plans that include risk assessments, identification of hazards near and/or around the activity venue, assessment of tidal and current patterns as well as weather conditions. In the event of unfavourable weather, dive leaders/instructors/trainers shall reschedule the dive for another day. Dive plans shall also incorporate emergency evacuation plans, specifying emergency contact numbers and access to the nearest emergency medical facility.
- d) Conduct safety briefings, detailing information including, but not limited to, the following:
 - Objectives of dives;
 - The buddy system, including what to do in the event of a lost buddy; and
 - Safety procedures such as descent and ascent, confined waters emergency drills, coping with strong currents, and topside supervision, dive site, dive procedure, estimated dive time, emergency procedure and unusual hazards or environmental conditions.
- e) Supervise participants closely and in full view of the entire group during descent, ascent and during training/leisure activities.
- f) Have on hand all relevant details of the activity – including all training dives, student records, medical declarations and liability releases – in the operations logbook and the divers' logbooks.
- g) Provide essential diving equipment during the dive activity, including but not limited to the following:
 - Timing devices;
 - Depth and pressure gauges;
 - Dive tables or dive computers;

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- Oxygen equipment capable of providing near 100 % oxygen to a breathing diver and at least 50 % oxygen to a non-breathing diver until medical assistance is available;
 - First-aid supplies;
 - Diver's flag;
 - AED;
 - Sound signalling system;
 - Floatation spine board;
 - Marine radio;
 - Mobile phone or satellite phone if diving away from mobile and radio access;
 - Searchlight (for night diving);
 - Emergency flares or smoke signals; and
 - Vinegar for box-type jellyfish stings.
- h) Conduct debriefs after activities and log all relevant information, and
- i) Report all diving accidents and incidents to the operator.

O.5.2 Certified dive leaders/instructors/trainers

O.5.2.1 A certified dive leader, instructor or trainer is a person that has been verified by a certified examiner to have met the necessary requirements. This certification is valid for a lifetime and typically renewed annually, which includes meeting the criteria specified by the certified examiner's training agency.

O.5.2.2 Certified dive leaders, instructors and trainers shall comply with the requirements outlined in this document, even when operating from overseas dive sites or using facilities provided by overseas partners and suppliers. When foreign dive instructors conduct dives at overseas dive sites, local service providers shall ensure they comply with the requirements outlined in this standard.

O.5.2.3 Certified dive leaders/instructors/trainers shall provide evidence that they are medically fit to dive.

O.5.3 Medical support and first aid

O.5.3.1 Dive leaders, instructors and trainers shall possess valid certifications in CPR/AED and first aid from a recognised certification body.

O.5.3.2 Dive leaders, instructors and trainers shall ensure that the oxygen cylinders they use have been obtained from a recognised certification body and are valid.

O.6 Service providers' responsibilities

O.6.1 Service providers shall have responsibilities including, but not limited to, the following:

- a) Ensuring the presence of at least one certified and experienced dive master, dive leader, instructor or trainer at the dive site during diving activities. This person shall have valid qualifications for their role, and competent supervision capabilities.

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- b) Providing a comprehensive dive plan, including emergency and contingency plans, prepared by the dive master, dive leader and instructor.
- c) Ensuring that all dive masters, dive leaders and instructors have valid CPR and AED certificates and are capable of providing necessary medical support for diving emergencies.
- d) Appointing an accredited oxygen cylinder vendor.
- e) Equipping each dive trip with at least one set of oxygen resuscitator and AED.
- f) Performing regular servicing of medical equipment in accordance with the vendor's recommendations.
- g) Communicating the latest medical response plan for the dive activity to all participants.
- h) Reporting of all accidents and incidents by the dive masters or dive leaders and instructors.
- i) Ensuring the vessel's seaworthiness and fitness for purpose. It is presupposed that vessels operating within the Port of Singapore are in compliance with applicable statutory and regulatory requirements, and
- j) Maintaining all required documentation, including participants' records, medical declarations, and liability releases, in proper order retaining them for a minimum period of 3 years.

O.6.2 Equipment

Service providers shall ensure the following:

- a) Diving and dive-related equipment available for rent are reliable and are of tested brands; and
- b) Regulators, masks, and snorkels available for rent are thoroughly cleaned and sterilised after each use by a competent individual.

O.6.3 Inspection, testing and certification

O.6.3.1 Service providers shall be responsible for ensuring that all diving and dive-related equipment supplied to participants for rental are in good working condition.

O.6.3.2 Participants shall be responsible to carry out final checks on the rental equipment before use.

O.6.3.3 Dive leaders, instructors, trainers and participants shall ensure, when taking over the gas cylinders that:

- a) only compressed air is used;
- b) maximum gas pressure does not exceed 200 bar;
- c) cylinders are stored under cover and in a well-ventilated area;
- d) all cylinder valves are serviced;
- e) gas is odourless and taste-free; and
- f) submersible pressure gauges used with cylinders provides accurate readings.

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O.6.4 Maintenance

The maintenance requirements for dive-related equipment are as follows:

- a) Scuba cylinder valves shall be tested by certified technicians at intervals not exceeding 12 months. 'O' rings shall be checked for wear and splitting before each dive.
- b) Gauges shall be inspected, calibrated, and tested by certified technicians before first use and every 12 months thereafter.
- c) Personal floatation systems, buoyancy compensators, dry suits, or other variable volume buoyancy compensation devices shall be equipped with an exhaust valve (over-pressure relief valve). These devices shall be functionally inspected and tested at intervals not exceeding 12 months. The depth gauge shall:
 - have an accuracy of within 1 % of the maximum scale reading for measuring depths less than 30 m; and
 - in other cases, have an accuracy of within 2.5 % of the actual condition it is measuring.
- d) Service providers shall establish a schedule of planned maintenance for all dive and dive-related equipment. The schedule shall identify the frequency at which each unit of equipment is to be checked and serviced, as well as the individual responsible for completing this task. The individual involved shall maintain a record of work done, either in hard or soft copy.

O.7 Buddy system

O.7.1 The buddy system shall be adopted for all dives. It involves minimally two participants (i.e. the buddies) to operate together as a single unit so that they can monitor and help each other throughout their dive activity. Hence, a diving group with odd numbers should be avoided, where possible.

O.7.2 The buddy system is effective in mitigating diving incidents, including, but are not limited to the following:

- a) Out-of-air emergencies;
- b) Non-diving medical emergencies;
- c) Entrapment in ropes or nets; and
- d) Avoidance of omission, misuse and failure of diving equipment.

O.7.3 Each participant should be responsible for the other participant, i.e. their assigned buddy. Buddies should:

- a) look out for each other and ensure no one is left behind; in the event of a buddy becoming unconscious, execute an unconscious diver rescue.
- b) adhere to safety protocols and follow the agreed-upon plan; and
- c) maintain physical proximity to each other to spot potential problems and respond quickly if an emergency arises, and dive safely.

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Annex P (normative)

Snorkelling

P.1 General

P.1.1 Snorkelling can be enjoyed in open waters and near beaches where there are beautiful coral reefs and fish. Snorkelling equipment, which are readily available for purchase, typically includes a pair of fins, a snorkel and mask.

P.1.2 Beginners to snorkelling shall familiarise themselves with snorkelling equipment, understand how they work and seek advice on potential risks and hazards, especially when snorkelling in open waters. It is risky for untrained snorkelers to venture into open waters for recreational snorkelling on their own.

P.2 Personal responsibility

P.2.1 Participants new to snorkelling shall undergo training (such as a certified proficiency snorkelling course) and be well-versed with use of equipment, risks and dangers of underwater environment, tidal and wave conditions, weather conditions and safety procedures.

P.2.2 Participants shall practise the following:

- a) Always snorkel under the supervision of qualified and experienced snorkelling leaders to ensure that activities are conducted safely.
- b) Be proficient in emergency rescue drills and be prepared to execute them if needed.
- c) Never snorkel alone and, instead, use the buddy system. Participants shall adopt the “one up, one down” practice. Participants shall care for buddies and always observe and detect for “silent vulnerable” participants who are likely to freeze or choke. Participants shall practise and establish good hand signals and communication.
- d) Always wear serviceable and well-fitted PFDs, that match their body size and weight, as per manufacturer’s specifications. Participants shall always ensure that the weight-buoyancy ratio is appropriate.
- e) Always stay near coastlines and avoid venturing too far into open waters. If participants are carried away by currents, it is crucial to stay calm and not panic. Drift along with currents, refrain from swimming against them, and conserve energy. Practise the bobbing breathing technique and use whistle signals or distress arm/hand signals to attract the attention of passing boats for assistance.
- f) Be attuned to their own physical condition and if feeling tired or breathless, practise the “stop, think, breathe, regain composure” sequence, remove masks, rest and breathe in fresh air. It is advisable to take a break after every 20 min of snorkelling. Additionally, participants should be aware of the risks of shallow water blackout (refer to Annex F).

P.2.3 In the event of inclement weather, participants shall stay together to ride out storms and provide mutual support. Participants shall leave no one behind and stay together as a group. In high wave situations, participants shall use their snorkels at the surface of the water. To conserve energy, participants shall relax and execute the bobbing technique of breathing on the surface. Participants shall not swim against the current; instead, employ the “stop, think, breathe, regain composure” sequence and huddle together, and not panic but take control of the situation.

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P.3 Responsibilities of snorkelling leaders/instructors/trainers

Leaders/instructors/trainers shall do the following:

- a) Be knowledgeable, competent, and experienced in open water conditions as well as in the training/leisure activities that they are providing. Their primary responsibility shall be ensuring the safety of training/leisure activities. Their secondary responsibility shall be selecting suitable, safe open water environments for these activities.
- b) Maintain a small coach/trainer-trainee ratio of 1:4.
- c) Understand trainees' profiles and swimming capabilities/abilities:

- Ensure trainees complete pre-participation questionnaire;

NOTE – Refer to the Sport Singapore website for an example of a pre-participation questionnaire.

- Conduct open water orientation programmes and provide guidance on the use of safety equipment and what to do during emergencies;
- Group trainees with similar capabilities together; and
- Plan regular breaks during activities and avoid venturing too far out into open waters.

Develop comprehensive plans for snorkelling trips, including risk assessment plans, identification of hazards/dangers around activities venues, tidal/current flow patterns, weather conditions and emergency plans. In the event of inclement weather, leaders, instructors and trainers shall reschedule snorkelling trips for another day.

- d) Conduct safety briefings that outline activity objectives and safety procedures. Debrief sessions shall be conducted by the leader, instructor or trainer after the activities conclude, and
- e) Supervise trainees closely and maintain full visibility of the groups during activities.

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Annex Q (normative)

Stand-up paddleboarding

Q.1 General

Q.1.1 Stand-up paddleboarding is a water sport derived from surfing. Stand-up paddle boarders stand on boards that float on waters and use paddles to propel themselves through the water.

Q.1.2 Beginners should use wider boards to ensure better stability. Narrower boards are more agile and are typically used by advanced paddlers or for stand-up paddle surfing.

Q.2 Personal responsibility

Prior to the commencement of the activity, participants shall do the following:

- a) Receive a briefing from the coach/trainer and organiser on their personal responsibilities;
- b) Understand and acknowledge their own abilities and limitations, and engage in activities within their skill levels;
- c) Be proficient in swimming;
- d) Avoid paddling alone. They shall paddle in groups to ensure assistance can be provided during emergencies; and
- e) Inform someone about the location and expected duration of activities.

Q.3 Weather and water conditions

With regard to weather and water conditions, participants shall do the following:

- a) Check weather forecasts and avoid paddling during inclement weather.
- b) Understand local and tidal conditions. A good knowledge of surrounding environment helps prepare for any eventualities. Participants shall practise and be well-versed in falling away from boards and climbing back on again. Participants shall also have knowledge of addressing common injuries, such as cramping, while in the water.
- c) Avoid entering the water when conditions are too demanding.

Q.4 Equipment and PFDs

Before embarking on activities, participants shall do the following:

- a) Check that the boards (either hard or inflatable) are in good condition. Inflatable boards shall be pumped up to the recommended pressure as determined by their manufacturer.
- b) Familiarise themselves with the weight, length, and thickness of the boards and be comfortable with them.

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- c) Ensure that any other equipment used is suitable for the activities and is in good working condition.
- d) Launch and recover within the relevant lifeguarded zones (black and white flags if on the coast).
- e) Wear appropriate attire (e.g. bright colours attires, for visibility).
- f) Wear appropriate PFDs with sufficient buoyancy, regardless of swimming abilities, to keep their body afloat even when unconscious.
- g) Wear suitable child-sized PFDs (for children under 13 years old) and be supervised closely, and
- h) Wear appropriate leashes (coiled or straight) that attach their ankle or calf to the board. Leashes are usually 30 cm longer than the board and serve to keep the board close to the participant, even if physically separated. Use ankle leashes for open or non-flowing waters and quick release waist-belt leashes for flowing water or tidal waterways.

Q.5 During paddling

During paddling, participants shall do the following:

- a) Be mindful of their surroundings. Activities shall be conducted away from motorised boats and swimmers. Participants shall always paddle in designated zones.
- b) Take turns and share waves, avoiding paddling through the line-up.
- c) Do not abandon their boards.
- d) Allow themselves to drift and reach the coast a little further away when caught by currents.
- e) Float on their backs to call for help if they fall into the water and lose their boards.
- f) Ensure they are not weighed down when they fall into the water, and
- g) Stay on top of their boards and signal for help if they encounter difficulties.

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Annex R (normative)

Water skiing and Wakeboarding

R.1 General

R.1.1 Water skiing is a surface water sport in which an individual is pulled behind a boat or a cable ski installation over a body of water, skimming the surface on two skis or one ski.

R.1.2 Wakeboarding is a surface water sport that involves riding a wakeboard over the surface of a body of water. It was developed from a combination of water skiing, snowboarding, and surfing techniques.

NOTE – Refer to the Sport Singapore website for the safe water-skiing and wakeboarding guide.

R.2 Personal responsibility

R.2.1 Participants shall do the following:

- a) Engage activities within their body's limits, abilities, and capabilities;
- b) Warm up before activities and cool down after activities;
- c) Water ski and wakeboard only at designated venues, avoiding obstacles and dangerous areas with rocks, strong currents, and sea lanes;
- d) Seek medical treatment at the nearest hospital or clinic when encountering sea creatures; and
- e) Ensure they have sufficient rest to make good judgements and decisions during activities.

R.2.2 The types of personal protection equipment (refer to Figure R.1) include the following:

- a) PFDs

These shall be worn when operating boats and doing activities. PFDs shall have a high impact rating and be designed for use for all water ski activities (except for tricks or show skiing, where the boat driver and observer shall be more observant to ensure the safety of the skier).

- b) Gloves

It is recommended to wear gloves for slalom and jump activities as they help protect the hands and improve grip.

- c) Helmets

These shall be worn for jump and wakeboard activities, especially if hitting jump ramps or any floating obstacles.

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Figure R.1 – Personal protection equipment

R.2.3 Before embarking on activities, boat operators, observers and participants shall do the following:

- a) Be able to swim;
- b) Check all equipment related to activities to ensure they are in good condition;
- c) Check the tide and weather forecasts;
- d) Avoid skiing when it is getting dark or at night, as it poses difficulties for boat operators, skiers and other boats to see clearly;
- e) Refrain from using drugs or alcohol before or during boat operation or water skiing, as these substances impair good judgment and coordination;
- f) Consult with physicians if they have any pre-existing medical conditions before participating in activities;
- g) Refrain from engaging in activities if feeling unwell, carrying existing injuries, or under medication;
- h) Participate in activities within their limits and not to the point of excessive fatigue;
- i) Bring snacks and water along for long sessions;
- j) Familiarise themselves with standard hand signals (refer to Figure R.2) and what to do in the event of an emergency; and
- k) Avoid skiing within 90 m of another vessel or 10 m of the shore, docks or swim areas.

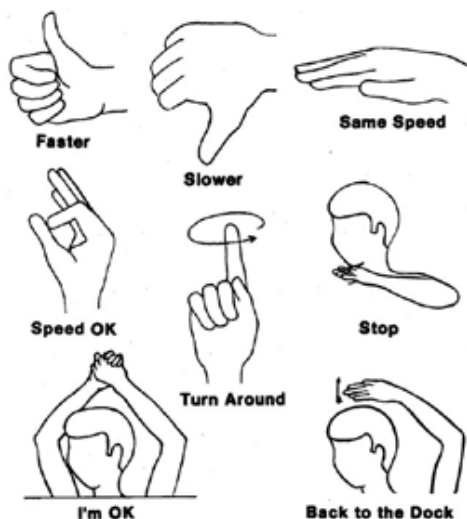


Figure R.2 – Hand signals

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R.3 Boat operators' responsibilities

R.3.1 Boat operators shall do the following:

- a) Have a valid licence to operate pleasure crafts in Singapore waters.
- b) Ensure the safety of skiers/riders by establishing and communicating rules to participants and passengers before hitting the water. They shall review the hand signals for water skiing/wakeboarding, plan the activity location once skiers/riders are behind boats and discuss expectations of everyone on board.
- c) Discuss ahead of time how they intend to pick up downed skiers and provide instructions how to properly re-board. Skiers shall also be briefed what to do in event of a fall.
- d) Avoid leaving boats running while skiers are attempting to re-board and never use engines or propellers as steps, and
- e) Give a wide berth to fishermen and slower-moving crafts (e.g. canoes, kayaks, and sail boats) as well as other water skiers being towed behind boats or present in the water.

R.3.2 Boat operators should be familiar with water-skiing areas so as to avoid shallow waters, submerged obstructions, and other dangerous situations. They should also stay clear of beaches and swimming areas, and refrain from intentionally spraying other boats or swimmers.

R.4 Observers' responsibilities

Observers on board boats shall do the following:

- a) Monitor skiers, receive signals and alert boat operators if skiers fall;
- b) Keep a lookout for boat traffic and potential obstacles in the water;
- c) Properly attach the towline to the boat's tower or pylon;
- d) Throw wakeboard/ski handles and lines to wakeboarders/skiers as needed;
- e) Retrieve wakeboard handles and line from the water after each session;
- f) Help balance the weight distribution in the boat for stability;
- g) Assist wakeboarders/skiers, especially beginners, with their equipment;
- h) Report wakeboarders'/skiers' signals and their current status to the boat operator; and
- i) Help and rescue injured wakeboarders/skiers in case of emergencies.

R.5 During water skiing and wakeboarding

R.5.1 All activities shall cease in the event of lightning, storms, or heavy rain, or if thunderstorms are approaching.

R.5.2 When two participants are water skiing/wakeboarding simultaneously behind the same boat, always use ropes of equal lengths.

NOTE – The use of equal lengths is not always necessary as it depends on the activity (e.g. show skiing).

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R.5.3 During water skiing and wakeboarding, participants shall do the following:

- a) Only indicate to boat operators to accelerate, with signal “Hit it!” only when water ski/wakeboard lines are tight with no slack.
- b) Not put any part of their bodies through handles or wrap the tow ropes around their bodies while water skiing or wakeboarding. If participants fall, they shall never try to grab the dragging handles/ropes.
- c) Look out for fallen skiers or riders. They shall leave as much space as possible between themselves and fallen skiers/riders when passing. To avoid collision, participants shall let go of the ropes and allow themselves to fall. After falling, participants shall look around to see if other skiers/riders or ropes are approaching, then swim to shore at once, taking equipment along unless during an emergency. Participants shall swim back to shore if cable system stops.
- d) Avoid fending off dragging handles; instead, they shall duck underwater to allow handles to pass over them.
- e) Avoid water skiing or wakeboarding directly in front or in the path of other boats or water skiers or wakeboarders.
- f) Avoid fast landings on shore. Participants shall not approach or hit any obstacles if landing areas are not clear of other skiers/riders. If obstacles cannot be avoided or are too close to the shore, let go of the ropes and fall into the water.
- g) If participants fall in an area where there is other boat traffic, they shall alert other boaters of their position by lifting more than half their water ski or wakeboard out of the water.

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